



A comprehensive survey: Whale Optimization Algorithm and its applications

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ABSTRACT

Whale Optimization Algorithm (WOA) is an optimization algorithm developed by Mirjalili and Lewis in 2016. An overview of WOA is described in this paper, rooted from the bubble-net hunting strategy, besides an overview of WOA applications that are used to solve optimization problems in various categories. The best solution has been determined to make something as functional and effective as possible through the optimization process by minimizing or maximizing the parameters involved in the problems. Research and engineering attention have been paid to Meta-heuristics for purposes of decision-making given the growing complexity of models and the needs for quick decision making in the engineering. An updated review of research of WOA is provided in this paper for hybridization, improved, and variants. The categories included in the reviews are Engineering, Clustering, Classification, Robot Path, Image Processing, Networks, Task Scheduling, and other engineering applications. According to the reviewed literature, WOA is mostly used in the engineering area to solve optimization problems. Providing an overview and summarizing the review of WOA applications are the aims of this paper.

1. Introduction

In many optimization problems, it is necessary to find the optimal solution to a specific problem under highly complex constraints over a reasonable period of time [1]. Such kinds of optimization issues are generally dealt with by modern intelligent methods [2]. Various strategies are suggested to resolve these issues, yet they are not sufficient enough to obtain better results [3,4]. In the last years, the researchers gave a lot of attention to meta-heuristic optimization algorithms in scientific communities with significant developments, mainly for solving many complex optimization problems [5–7]. Hill-Climbing, Random Search and Simulated Annealing [8], were previously regarded as the traditional algorithms to solve the optimization problems before meta-heuristic algorithms [9]. The search of Traditional algorithms is started from one point and gradient information is required that took a while to reach the global optima [10]. As regards their constrained relevance and complexity of limitations, these algorithms were not very effective in solving life-like applications such as localization issue [11], economical optimization [12], structural optimization problems [13] and engineering design problems [14] engaging different limitations to be gratified.

Meta-heuristic algorithms have performed in a promising manner to

solve most real-world optimization problems being highly nonlinear and multimodal [15]. A certain tradeoff of randomization and local search is used by all meta-heuristic algorithms. Good solutions can be found by these algorithms for difficult optimization problems, but it is not guaranteed that optimal solutions could be reached. We hope that these algorithms work most of the time, but not all the time. Meta-heuristic algorithms could be convenient for global optimization [16]. Meta-heuristics are generally categorized into two classifications: single-solution and multiple-solution based problems [17].

The searching process begins with one candidate solution in single-solution based algorithms while the optimization is performed using a set of solutions (i.e., population) in multiple-solution based algorithm [18]. Multiple-solution or population based meta-heuristics are advantageous as compared to single-solution based metaheuristics. The advantages are as follows [18]:

- The searching process commences with random produced population, namely a group of multiple solutions.
- The information could be shared by the multiple solutions between each other around the search space and local optimal solutions could be prevented.

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- Multiple-solution or population based metaheuristics are more able to explore better than the single-solution based techniques.

Moreover, algorithms that are based on meta-heuristic Multiple-solution are divided into three classes like evolutionary-based, physics-based, and swarm-based methods [18]. The first division is generic population based meta-heuristic rooted from biological evolution that are recombination, mutation, and finally selection. There are no assumptions for these methods as regards the basic fitness landscape. Genetic Algorithm (GA) [19], that is a category of well-known optimization methods, uses the theory of Darwinian evolution. Biogeography-based optimization (BBO) [20] algorithm is a new kind of optimization technique based on biogeography concept. Differential evolution (DE) [186–188] is one of the most popular and efficient evolutionary algorithms for numerical optimization and it have gained much success in a series of benchmark functions as well as real applications.

Algorithms that are Physics-based are the second group of algorithms. In this type, each search agent can interact and move at the search space in accordance with some physics rules. The rules of gravitational force, inertia force, electromagnetic force, and the others can be mentioned. As an example, Simulated Annealing (SA) [8] was inspired by the metal-lurgic annealing process, Ray Optimization (RO) by the Snell's light refraction law [21], Electromagnetic Field Optimization (EFO) [22], and Gravitational Search Algorithm (GSA) [23] by the law of gravity and mass interactions.

The last division is swarm-based algorithms driven from the collective behavior of social creatures [24]. This collective intelligence is based on swarm interactions with each other. It is easier to implement swarm-based algorithms than the evolutionary based algorithms because of including the lower number of operators (i.e., selection, crossover, mutation). Furthermore, swarm-based algorithms have some advantages examples of which are as follows:

- The information about the search space can be maintained by swarm-based algorithms at the time of iterations whereas the information of the previous generations can be maintained by evolutionary-based algorithms can.
- The evolutionary techniques have more input parameters as compared to swarm-based algorithms.
- Memory space is less utilized by swarm-based algorithms so as to save the best optimal solutions.

Meta-heuristics are high quality methods guiding the search agents to gradually enhance the whole solution [24]. Henceforth, the optimization of the solution is carried out by checking across some random elements and possibilities, hoping to find a more qualified solution, whereas the candidate solution is inherited from one stochastic iteration to another. An absolute best solution might not be yielded by meta-heuristics as opposed to deterministic methods. However, more often than not they generate sufficiently good solution mainly during a reasonable period of time.

Biological or physical phenomenon can be fundamentally depersonalized by meta-heuristic algorithms to deal with intricate real world optimization problems. These meta-heuristic algorithms are mostly derivation-free unlike classical techniques [25]. Meta-heuristic algorithms are excellently able to avoid local optima entrapment given their stochastic nature [200]. Due to their simplicity, flexibility, robustness, and efficiency, such algorithms can be applied to various fields [25]. Particle Swarm Optimization (PSO) [26], Artificial Bee Colony (ABC) [27], Firefly Algorithm (FA) [28], Selfish Herd Optimizer (SHO) [29], Bat Algorithm (BA) [30], Krill-Herd (KH) [31], Cat Swarm Optimization [32], Grey Wolf Optimizer (GWO) [33], Ant Lion Optimizer (ALO) [34], Butterfly Optimization Algorithm (BOA) [35], Virus Colony Search (VCS) [36], Squirrel Search Algorithm (SSA) [37], Water Wave Optimization (WWO) [38], Galactic Swarm Optimization (GSO) [39], Vortex Search (VS) [40], Crow Search Algorithm (CSA) [41], and most recently WOA

[42] are examples of the most important nature driven meta-heuristic algorithms developed so far.

Nature-inspired optimization algorithms, including evolutionary algorithms (EAs), Physics-based, and swarm-based algorithms (SAs), are part of the computer intelligence discipline that has become increasingly popular over the past few decades. The popularity of nature-inspired optimization algorithms is due to the fact that they can be used to solve complex problems in different domains [189]. Nature-inspired optimization algorithms, especially SAs, solve many scientific and engineering problems due to their flexibility and simplicity [190,191].

These algorithms are started with a group of randomly chosen candidate solutions of the optimization issue and attempt to enhance the quality of the group on the basis of a cost function. According to a series of simple mainly nature-inspired rules, the existing solutions are perturbed repeatedly to enhance their cost function values. The solution is found by the algorithms mentioned above through efficiently searching the search space and reducing its size. In fact, algorithms are mainly different from each other in their way of creating balance between exploration (global search capability) and exploitation (local search capability around the near-optimal solution) [192,193]. In addition, investigating a suitable balance between exploration and exploitation is the most difficult task in the process of development of any meta-heuristic algorithm because of stochastic nature of the optimization process [42]. The exploration phase aids the optimizer to explore the search space in a global manner as widely as possible. Furthermore, the population in this phase is challenging with changes that are abrupt, whereas, the exploitation phase involves improving the promising solutions accessed of the exploration phase.

Mirjalili and Lewis have recently developed a new optimization algorithm called WOA [42] to meta-heuristic algorithm. It is assumed that whales are the most intelligent animals containing motion. The specific hunting behavior of humpback whales motivated the WOA algorithm. More usual, the humpback whales like to hunt fishes that are swimming near the surface of sea such as krill's or small fishes. Humpback whales make use of their own specific unique hunting technique named bubble net feeding method. Using this method, the whales swim around the prey and make recognizable bubbles along a circle or '9'-shaped path as shown in Fig. 2.

The rest of the paper is organized as follows. Structure of WOA is generally described in Section 2 while WOA is reviewed in relation to its hybridization, improved, and variants, in Section 3. An overview of WOA in solving various engineering optimization problems is provided in Section 4. In Section 5, WOA is discussed and directions for further development of these algorithms are covered. Finally, the conclusion of the paper is provided in Section 6.

2. Whale Optimization Algorithm

WOA is a swarm intelligence algorithm proposed for continuous optimization problems. It is proven that this algorithm has better or comparable performance as compared to some existing algorithmic techniques [42]. The hunting behavior of the humpback whales has been the source of inspiration for WOA [42]. In WOA each solution is thought to be a whale. In this solution, a whale tries to replete a new place in the search space considered as a reference the best element of the group. Two mechanisms are used by the whales to be used for search for the prey location, and attack. In the first one, the preys are encircled and the second creates bubble nets. Regarding optimization, when the whales look for a prey the exploration of the search space is performed and the exploitation occurs during the attack behavior.

Many researches have been conducted on WOA. In this paper as a review paper, the materials have been collected by using the WOA as a keyword through two stages. Firstly, such creditable publishers as Elsevier, Springer, and IEEE as well as other journals which were searched using Google Scholar have covered the papers published about WOA.

Fig. 1 demonstrates the number of papers published by various researchers from 2016 to 2018 on WOA as well as its variants according to the different report databases such as Elsevier, Springer, IEEE, and more. It is suggested by Fig. 1 that in last two years publications of WOA research have enormously increased.

2.1. Inspiration

Whales are beautiful creatures; they are regarded as the biggest mammals in the world. 30 m long and 180 t weight is the range of weight that an adult whale can grow up to. *Killer, Minke, Sei, humpback, right, finback, and blue* are the various main species of giant mammals. Whales are mainly considered to be predators. Whales never sleep because they have to breathe from the surface of oceans. As a matter of fact, only half of whales' brain sleeps. What is interesting about the whales is that they are regarded as highly clever and emotional animals.

Hof and Gucht [43] believe that there are common cells in certain areas of whales' brains like those of human being. These areas are known as spindle cells. They are accountable for judgment, emotions, and social behaviors in humans. In other words, the spindle cells distinguish us from other creatures. The main cause of Whales smartness is the number of these cells which are more than the number of an adult human. It is a fact that whale can think, learn, judge, interact, and get even emotional as a human does, but clearly with a much lower level of smartness. It has been observed that whales (mostly killer whales) are able to develop their own dialect as well.

Another interesting fact about whales is their social behavior. Whales live alone or in communities. However, they are mainly observed living in groups. Some of whales species (killer whales for instance) can live in a family over their entire life period. One of the biggest baleen whales is humpback whales (*Megaptera novaeangliae*). An adult humpback whale is approximately the size of a school bus. What they like to hunt are krill and small fish herds.

Whales' special hunting strategy is the most intriguing quality of the humpback whales. Their hunting behavior is known as bubble-net feeding method [44]. Humpback whales prefer to hunt school of krill or small fishes near the surface. Before 2011, this behavior was only investigated according to the surface observation. However, in Ref. [45] it has been surveyed that this behavior utilizes tag sensors. Out of 9 individual humpback whales 300 tag-derived bubble-net feeding events were caught. Two maneuvers associated with bubble were found and named 'upward-spirals' and 'double-loops'. Humpback whales dive around 12 m down in the former maneuver and then start making bubble in a spiral shape around the prey and swim up toward the surface. The latter consists of three different stages: coral loop, lob-tail, and capture loop. You can find detailed information about these behaviors in

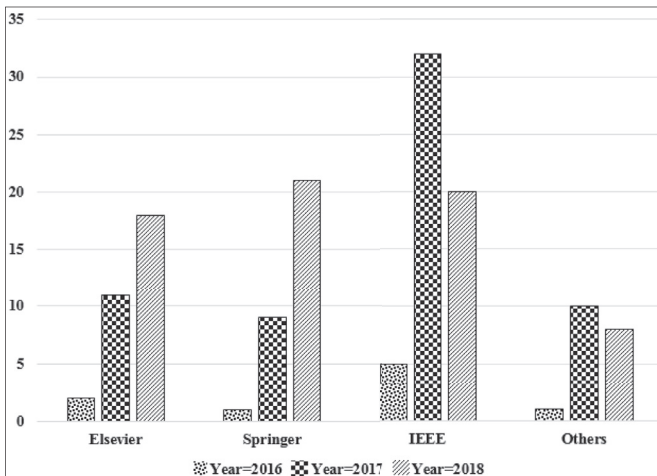


Fig. 1. Trends of WOA application from year 2016–2018.

Ref. [45]. It is worth noting here that bubble-net feeding is a unique behavior that can merely be observed in humpback whales.

2.2. A mathematical-based model and optimization algorithm

In this section, a model that is mathematical-oriented, at the first phase, is provided from encircling prey, search for prey and spiral bubble-net feeding maneuvers.

2.2.1. Encircling prey

Humpback whales can recognize and cover the location of prey. As it is not possible to know the position of the optimal design in the search space a priori, it is assumed by the WOA that the target prey is the best candidate solution or is near to the optimum. Once the best search agent is elaborated on and is defined, the next phase can be the attempt of the other search agents in order to update their positions towards the best search agent. This behavior is shown in the form of the equations as follows [42]:

$$\vec{D} = \left| \vec{C} \cdot \vec{X}^* (t) - \vec{X}(t) \right| \quad (1)$$

$$\vec{X}(t+1) = \vec{X}^* (t) - \vec{A} \cdot \vec{D} \quad (2)$$

In Eqs. (1) and (2) t represents the current iteration, $\vec{A}$ and $\vec{C}$ indicate coefficient vectors, $\vec{X}^*$ represents the position vector of the best solution that already has been obtained, $\vec{X}$ is the position vector, $||$ is the absolute value, and $\cdot$ is one by one element multiplication. It is worth noting here that $\vec{X}^*$ should be updated in each iteration if there is a better solution. The vectors $\vec{A}$ and $\vec{C}$ are calculated as follows [42]:

$$\vec{A} = 2\vec{a} \cdot \vec{r} - \vec{a} \quad (3)$$

$$\vec{C} = 2 \cdot \vec{r} \quad (4)$$

In Eq. (3) $\vec{a}$ is linearly decreased from 2 to 0 over the course of iterations (in both exploration and exploitation phases) and $\vec{r}$ is a random vector in $[0, 1]$. Fig. 3(a) demonstrates the rationale behind Eq. (2) for a 2D problem. The position (X, Y) of a search agent can be updated on the basis of the position of the current best record (X^*, Y^*). Various positions around the best agent can be achieved with respect to the current position by adjusting the value of $\vec{A}$ and $\vec{C}$ vectors. Fig. 3(b) shows the possible updating position of a search agent in 3D space. It is worth mentioning that by offering a definition of the random vector ($\vec{r}$) it is possible to reach any position in the search space located between the key-points shown in Fig. 3. Thus, in Eq. (2) any search agent is allowed to

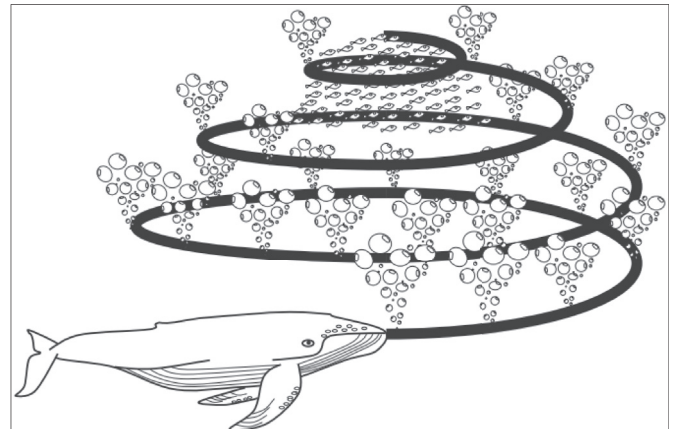


Fig. 2. Bubble-net feeding behavior of humpback whales [42].

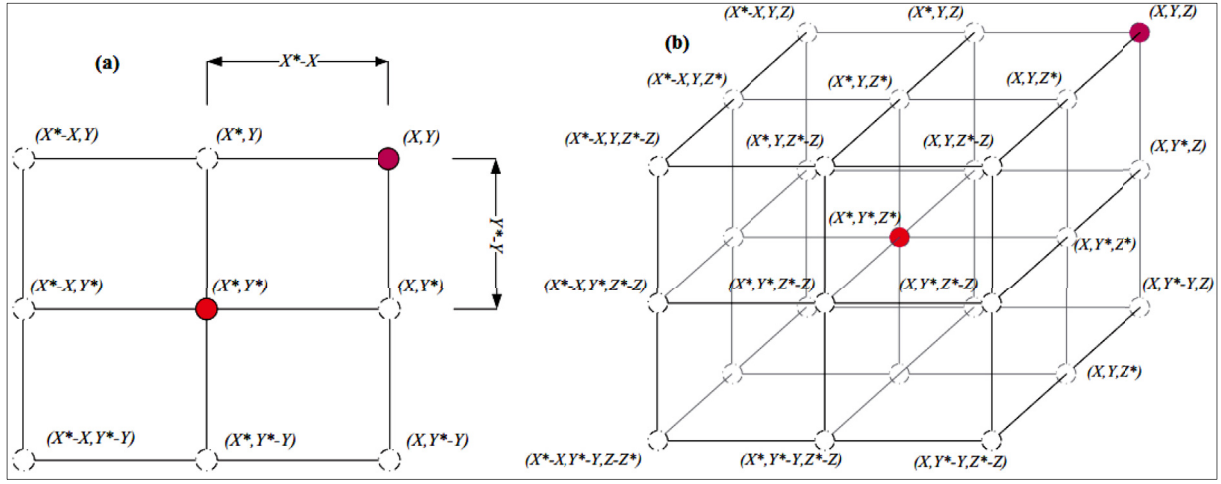


Fig. 3. 2D and 3D position vectors and their possible forthcoming locations (X^* represents the best solution obtained up to now) [42].

update its position near the current best solution and simulates encircling the prey.

It is possible to enlarge the same concept to a search space with n dimensions, and the search agents can move around the best solution in hyper-cubes achieved up to now. As mentioned earlier, the prey could be attacked by the humpback whales with the bubble-net strategy. The mathematic formulation of this method is as follows:

2.2.2. Bubble-net attacking method (exploitation phase)

Two approaches have been made to create a mathematical model of the bubble-net behavior of humpback whales. The two approaches are as given below:

- 1) Shrinking encircling mechanism: the value of $\vec{a}$ in Eq. (3) is lowered to achieve this behavior. Remember that the range of $\vec{A}$ fluctuation is also decreased by $\vec{a}$. To put it in another way, $\vec{A}$ represents a random value in the interval $[-a, a]$ in which a is lowered from 2 to 0 in the process of repetitions. While defining random values for $\vec{A}$ in $[-1, 1]$, the new position of a search agent can be defined somewhere between the original position of the agent and the position of the current best agent. Fig. 4(a) illustrates the probable positions from (X, Y) towards (X^*, Y^*) that can be achieved by $0 \leq A \leq 1$ in a 2D space.
- 2) Spiral updating position: As shown in Fig. 4(b), in this method, the distance between the whale located at (X, Y) and prey located at (X^*, Y^*) is first calculated. A spiral-based equation that is placed between the place of whale and prey to imitate the helix-shaped movement which is related to humpback whales is as follows:

$$\vec{X}(t+1) = \vec{D} \cdot e^{bl} \cdot \cos(2\pi l) + \vec{X}^*(t) \quad (5)$$

where $\vec{D} = |\vec{X}^*(t) - \vec{X}(t)|$ and shows the distance of the i^{th} whale to the prey (best solution obtained so far), b represents a constant for a definition of the logarithmic spiral shape, l is a random number in $[-1, 1]$, and $(.)$ is an element-by-element multiplication.

Remember that within a shortening circle and along a spiral-shaped path humpback whales swim around the prey simultaneously. In order to obtain a model of this simultaneous behavior, it is assumed that there is a 50% probability to select between either the shortening encircling mechanism or the spiral model to upgrade the position of whales in the process of optimization. The mathematical model is as follows [42]:

$$\vec{X}(t+1) = \begin{cases} \vec{X}^*(t) - \vec{A} \cdot \vec{D} & \text{if } p \leq 0.5 \\ \vec{D} \cdot e^{bl} \cdot \cos(2\pi l) + \vec{X}^*(t) & \text{if } p \geq 0.5 \end{cases} \quad (6)$$

In Eq. (6) p indicates a random number in $[0, 1]$. Next to the bubble-net method, the humpback whales search for prey randomly. Pay attention to what follows for the mathematical model of the search.

2.2.3. Search for prey (exploration phase)

$\vec{A}$ vector variation based approach can be made use of in searching for prey (exploration). As a matter of fact, according to the position of each other, humpback whales search randomly. Hence, $\vec{A}$ has been used with the random values more than 1 or less than -1 for mobilizing search agent so as to move far away from a reference whale. As opposed to the exploitation phase, a search agent position is upgraded in the exploration phase on the basis of a randomly selected search agent instead of the best search agent found so far. This mechanism and $|\vec{A}| > 1$ puts emphasis on exploration and the WOA algorithm is permitted to perform a global search. The mathematical model is as follows [42]:

$$\vec{D} = |\vec{C} \cdot \vec{X}_{rand} - \vec{X}| \quad (7)$$

$$\vec{X}(t+1) = \vec{X}_{rand} - \vec{A} \cdot \vec{D} \quad (8)$$

In Eq. (7) $\vec{X}_{rand}$ represents a random whale position vector selected from the current population. Some of the possible positions around a particular solution with $|\vec{A}| > 1$ are depicted in Fig. 5.

The WOA algorithm is started using a group of random solutions. At each iteration, search agents renew their positions with regards to either a randomly selected search agent or the best solution obtained so far. The a parameter is lowered from 2 to 0 to provide exploration and exploitation, respectively. A random search agent is selected as $|\vec{A}| > 1$, while the best solution is selected as $|\vec{A}| < 1$ to renew the position of the search agents. Given the value of p , WOA has the ability interchange between either a spiral or circular movement. Finally, the satisfaction of a termination criterion terminates the WOA algorithm. Algorithm 1 presents the pseudo code that belongs to WOA algorithm. From a theoretical point of view, because of including exploration/exploitation ability we can consider WOA a global optimizer. In addition, a search space is defined by the proposed hyper-cube mechanism in the neighborhood of the best solution; other search agents are permitted to exploit the current best record inside that domain. The WOA algorithm is allowed a smooth transition between exploration and exploitation by Adaptive variation of the search vector A : by decreasing A , some iteration is allocated to exploration ($|A| \geq 1$) and the rest is dedicated to exploitation ($|A| < 1$). Remarkably, merely two major internal parameters are included by WOA for adjustment (A and C).

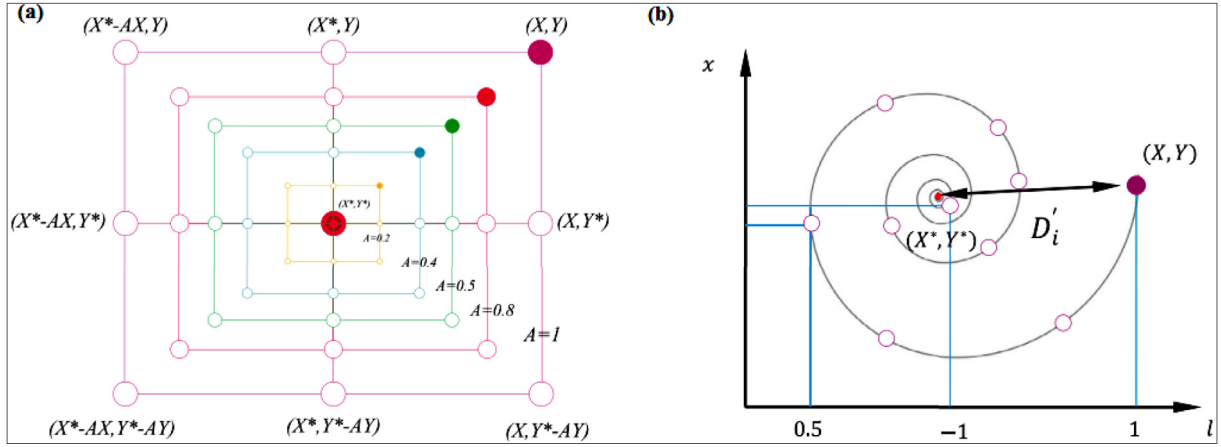


Fig. 4. Bubble-net search mechanism implemented in WOA (X^* is the best solution obtained so far): (a) shortening encircling mechanism and (b) spiral updating position [42].

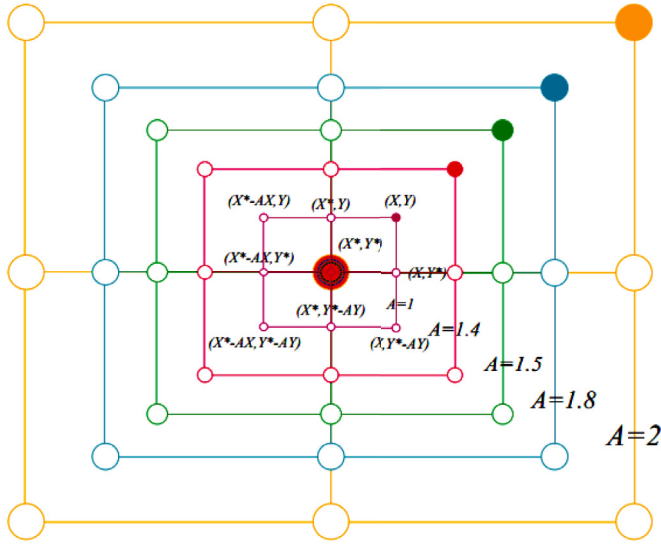


Fig. 5. Exploration mechanism implemented in WOA (X^* is a randomly chosen search agent) [42].

Algorithm 1

Pseudo-code of the WOA [42].

```

Initialize the whales population  $X_i$  ( $i = 1, 2, \dots, n$ )
Calculate the fitness of each search agent
 $X^*$  = the best search agent
while ( $t < \text{maximum number of iterations}$ )
  for each search agent
    Update  $a$ ,  $A$ ,  $C$ ,  $l$ , and  $p$ 
    if1 ( $p < 0.5$ )
      if2 ( $|A| < 1$ )
        Update the position of the current search agent by Eq. (1)
      else if2 ( $|A| \geq 1$ )
        Select a random search agent ( $X_{\text{rand}}$ )
        Update the position of the current search agent by Eq. (8)
      end if2
    else if1 ( $p \geq 0.5$ )
      Update the position of the current search by Eq. (5)
    end if1
  end for
  Check if any search agent goes beyond the search space and amend it
  Calculate the fitness of each search agent
  Update  $X^*$  if there is a better solution
   $t = t + 1$ 
end while
return  $X^*$ 

```

The flowchart of WOA is demonstrated in Fig. 6.

3. Various methods of WOA

Several methods of WOA algorithms exist in the literature. A certain categorization scheme is required for classifying them. Fig. 7 shows various methods of WOA.

3.1. Hybridization

Hybridization of two algorithms is a generic model to take the advantages of the two algorithms while lessening their impediments. The combined techniques have performed well for re-solving the specified problem, and the combination of these techniques can improve the results obtained on their own. Through hybridization of algorithms, we can enhance the exploration and exploitation of the algorithm completely [46,47]. Especially, lack of preciseness in an algorithm can be catered for by combination with a local search approach refining the results. The hybrid approach in the domain of optimization is growing more and more popular and improving the performance of classical algorithms on the basis of the notion of hybridizing the components from other optimization techniques is focused upon.

Traditionally, it is observed that the capability of an optimization algorithm to balance exploration and exploitation has a remarkable impact on its performance. Due to quick convergence, algorithm is more likely to get stuck in a local optimum. Given too much concentration on exploration near the end of the optimization run, time dedicated to further refine promising solutions would be wasted. Thanks to the significance of the effective administration of solution space diversity in traditional optimization algorithms, it is not a major stretch to think that the diversity of the heuristic space and how it is managed throughout the optimization run could have a significant role on performance of hybridization.

3.1.1. Meta-heuristic algorithms

Many scholars/researchers have been developing numerous hybrid nature-inspired approaches for the betterment of the exploration and exploitation performance of existing algorithms. The Hybrid Whale-PSO Algorithm (HWPSO), as a new population based hybrid meta-heuristic algorithm was proposed by Laskar et al. [48] to manage complex optimization problems. Due to excellent exploration capability of WOA algorithm, the proposed algorithm makes an attempt to conquer the limitations related in a PSO exploration phase that is Stagnation Effect, through combination with a WOA in a novel way. 'Forced' Whale and

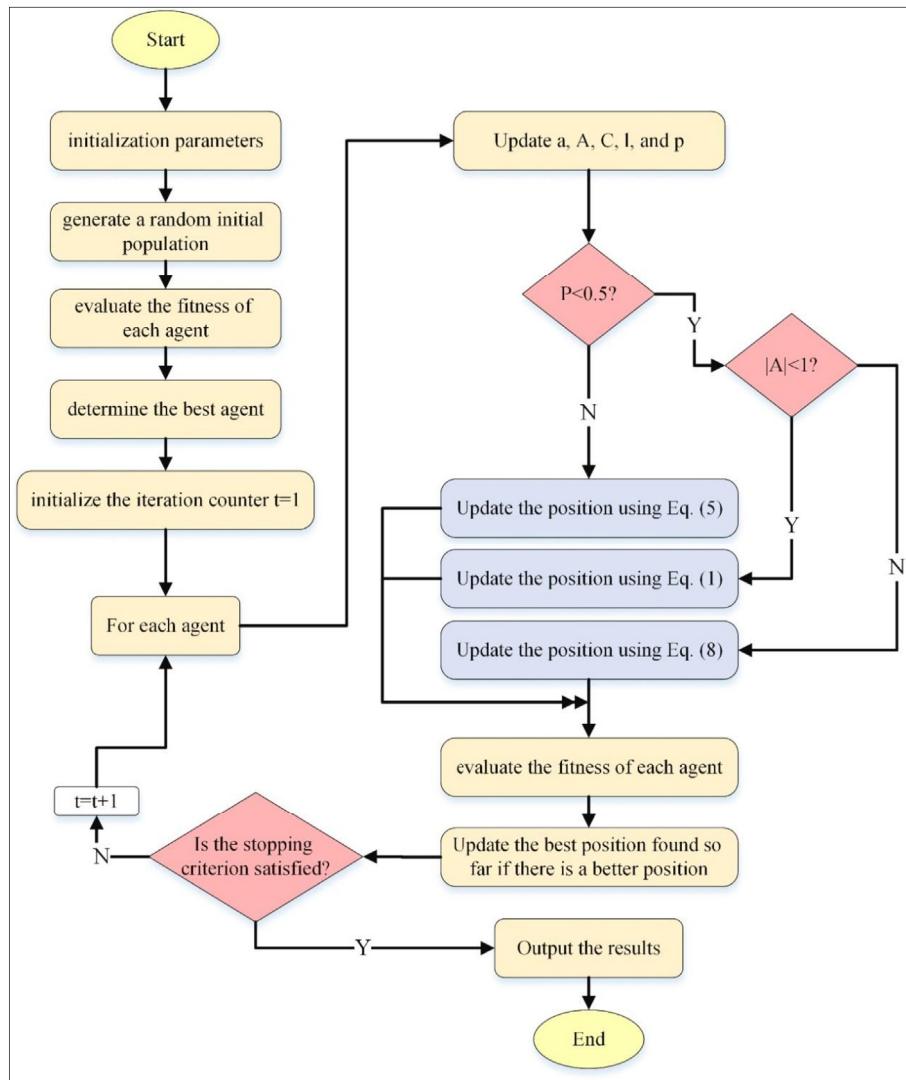


Fig. 6. The flowchart of WOA.

'Capping' Phenomenon are the two novel techniques utilized while hybridizing. In exploration phase, the 'Forced' WOA has been introduced and then it assists enables the WOA to lead PSO for better local optima avoidance and a new 'Capping' phenomenon that is simultaneously utilized while limiting the WOA search mechanism at the exploitation phase, hence it converges the solution to a global optimum value in more speed. The proposed HWPSO and its performance have been tested on benchmark mathematical functions and Electronics Design Optimization problems in which the numbers of the former was 18 and the number of the latter was 3. The simulation results are indicative of the fact that the proposed algorithm performed better than many state of the art algorithms by obtaining better optima with a very low standard deviation for most of the benchmark functions that have been used. The efficacy of this proposed algorithm that is HWPSO has been also examined by different analysis tools such as statistical as well as complexity analysis. Based on the results, it can be claimed that the performance of HWPSO algorithm seems to be satisfactory. The performance is efficient in all three cases of electronics design optimization problems, further validated with standard design instruments and the findings showed that they are in close agreement with each other.

A new technique hybrid PSO-WOA has been proposed by Trivedi et al. on some unconstraint benchmark test functions [49]. Hybrid PSO-WOA is the result of combining PSO that it is used for exploitation phase. Furthermore, WOA is used for exploration phase in uncertain

environment. The flaw of PSO is its inability to cover small search space at the time of solving higher order or complex design problem with the existence of constant inertia weight. Because of using logarithmic spiral function, WOA is used for exploration phase which consequently covers a vast area in uncertain search space. So WOA directs the particles faster toward optimal value and reduces the time of computation. Analyzing competitive results obtained from PSO-WOA are signs of efficacy validation in comparison with standard PSO and WOA algorithm.

Abdel-Basset et al. [50] proposed a new algorithm which combines the WOA with a local search strategy to defeat the permutation flow shop scheduling problem. The Largest Rank Value (LRV) is required by the algorithm to agree with the discrete search space of the problem. The candidate schedules are in a diversity mood by using a swap mutation operation as well. Furthermore, the insert-reversed block operation is designed in the form that to be escaped from the local optima. The proposed hybrid whale algorithm (HWA) is included with Nawaz-Enscore-Ham (NEH) in order to increase the performance of the algorithm. Local search strategy, swap mutation, insert-reversed block operation, and combining a heuristic algorithm called NEH HWA are the different techniques to converge toward optimal solutions in a short time. HWA was evaluated on four various benchmarks that are Carlier, Heller, Reeves, and Taillard benchmarks. The findings obtained from the computation are signs of the advantages and plus points of HWA.

Singh and Hachimi [51] proposed a hybrid nature-inspired

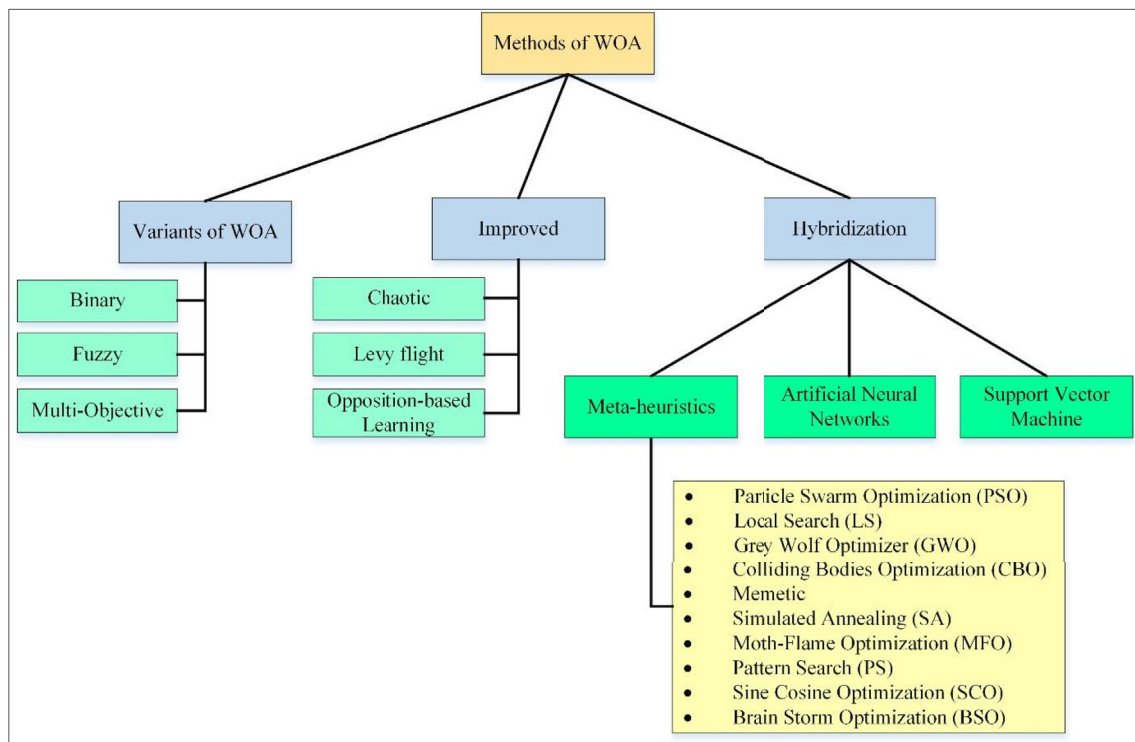


Fig. 7. Categorization of various methods of WOA.

optimization technique. The hybrid algorithm has been created utilizing Mean Grey Wolf Optimizer (MGWO) and WOA. They have used the spiral equation of WOA for two approaches in the Hybrid Approach GWO (HAGWO) algorithm: (I) at first place, the spiral equation has been utilized in GWO algorithm to make a balance between two elements of the exploitation and the exploration processes in the new hybrid approach; and (II) at the second place, this equation has also been used in the whole population to block the premature convergence and trapping in local minima. The WOA is used in the phase of exploration because of using logarithmic spiral problems, so it embeds vast areas in uncertain search spaces. Because of the fact that both of the variants are randomization approaches, we use unknown term search space while estimating over the course of iteration from the start till maximum iteration limit. Exploration phase is equal to the ability of the variant to examine large quantities of feasible solutions. Position of grey wolf that is likely to find the global optimum solution of the problem substituted with the position of whale equal to the position of grey wolf but highly influential to move a solution towards an optimal one. The wolves are directed to an optimal value and computational time is reduced. We know that GWO is a recognized approach that exploits the best possible solution from its unknown search space. Therefore, mixture of best characteristic (exploitation with Mean GWO and exploration with WOA) is a guarantee of obtaining best possible global optimal solution related to the real life and standard problems that also prevent local stagnation or local optima problems. The benefit of both Mean GWO in exploitation and WOA is merged by Hybrid WOA-MGWO in the exploration phase towards the targeted optimum solution. The feasibility and effectiveness of the hybrid algorithm have been examined by solving such standard benchmarks as XOR, Baloon, Iris, Breast Cancer, Welded Beam Design, Pressure Vessel Design problems comparing the results with those obtained through other metaheuristics. The solutions are a proof for the fact that the newly existing hybrid variant has higher stronger stability, faster convergence rate, and computational accuracy than other nature-inspired metaheuristics on the maximum number of problems and can successfully solve the function of constrained nonlinear optimization in reality.

In Ref. [52], through combining WOA with some concepts of the

Colliding Bodies Optimization (CBO) [53] attempts have been made to enhance solution accuracy, reliability, and convergence speed for the betterment of its original formulation. The new method, known as WOA-CBO algorithm, is applied to building site layout planning problem. Three case studies are chosen to demonstrate efficacy and performance of the WOA and WOA-CBO in building site layout problems. The first case is a discrete and equal area facility layout problem that can be assigned to any location by any facility; the second case is an unequal area version of discrete facility layout problem with more constraints; and the last case is a continuous model of building site layouts. These studies are investigated by WOA, CBO, and WOA-CBO, and the obtained consequences are put to comparison with each other. The results of this experiment confirmed the fact that the proposed approach could be applied for construction site layout problem. Furthermore, comparing WOA and WOA-CBO demonstrated that the proposed modification was efficient and obtained better solution with more stability.

In Ref. [54], a memetic WOA (MWOA) has been proposed through incorporation of a chaotic local search into WOA to enhance its exploitation ability. We expect that the global exploration and local exploitation could be well balanced by MWOA at the time of the search process, hence a better search performance. Verification of MWOA efficiency is actualized by forty eight benchmark functions. When it comes to the convergence speed and the solution accuracy experimental findings represented that the performance of MWOA is better than its competitors.

In Ref. [55], two hybridization models have been made utilize of in planning different feature selection techniques based on WOA. In the beginning model as the first one, SA algorithm is contained in WOA algorithm, whereas it is utilized to move forward the best solution found after each cycle of WOA algorithm in the second model. Using SA here is aimed at enhancing the exploitation by searching the most promising regions located by WOA algorithm. The execution of the proposed approaches is assessed on 18 standard benchmark datasets from UCI repository and the results compared and contrasted with three famous wrapper feature selection methods applied in the literature. The exploratory comes about affirm the effectiveness of the proposed

approaches in upgrading the categorization precision compared to other wrapper-based calculations that needs the ability of WOA algorithm in tasks such as searching the feature space and choosing the foremost enlightening properties for classification tasks.

In Ref. [56], the issue of deciding the ideal thresholding in a case of multilevel thresholding for picture division has been considered as an optimization issue where the Otsu's strategy has been utilized as a fitness function. Hence, two unused swarm algorithms, specifically, the WOA and Moth-Flame Optimization (MFO) [57] have been proposed to illuminate this issue. The objective of each algorithm is to decide the best thresholds value that increases the Otsu's function. The exploratory comes about of the proposed algorithms have been contrasted and compared to algorithms such as Sine Cosine Algorithm, FASSO, Social Spiders Optimization (SSO), HS, and FA. The execution of each algorithm has been evaluated based on four measures. The measures are as follows:

The best fitness values, PSNR, SSIM, and execution time. It has been used eight benchmark images in all treatments. The findings revealed that all the algorithms obtained the same fitness function value at a threshold level equal two; this is due to the fact that the segmentation or division issue becomes simple and easy at this level of threshold.

At the three-levels of thresholding, the WOA and MFO algorithms performed better in comparison with the other algorithms on nearly all images, based on PSNR and SSIM; however, WOA performed better than the other that is MFO. For a better number of thresholds, the MFO algorithm performed more than WOA algorithm in all images, and regarding the fitness value, the performances of PSNR, and SSIM were high. All things considered, as respects time complexity, investigation of recorded execution time needed algorithm has represented that the MFO algorithm uses higher execution time in comparison with FA, WOA, and SSO.

In Ref. [58], the WGC (WOA-GWO) has been used for clustering data. The WGC algorithm estimates the optimal centroid for better performance of the data clustering according to the minimum fitness function. The minimum fitness function is executed in terms of three fitness constraints, specifically the inter-cluster distance, intra-cluster distance, and the density of the cluster. The fitness measure is measured as a minimum value that requires the minimum inter-cluster distance, the maximum intra-cluster distance, and the maximum cluster density. Upon calculation of the optimal centroid with the WGC algorithm, this optimal centroid is utilized to cluster the data points present within the data. The clustered data covers the optimally clustered data and supplies all the important data for the decision-making handle. Execution investigation utilizing the three datasets is demonstrative of the reality that the proposed WGC outperforms all the existing methods by obtaining a minimum value that is 1.463 for MSE and the maximum value of 0.9716 for fitness, 0.9695 for rand coefficient, and 0.8949 for accord coefficient.

In Ref. [59], a hybrid whale-wolf optimization (WGWO) method has been used to precisely unravel the issue of financial expedite. This method productively coordinating the mechanisms of WOA and GWO with crossover and mutation operators. The adequacy of the proposed method has been illustrated via comparison with six optimization methods. The list of compared optimization methods were as follows:

GWO, WOA, PSO, ABC, ALO, and Dragonfly Algorithm (DA). Two different test systems (6 and 10 generating units) are used to evaluate the performance of the proposed method. The findings demonstrated that the hybrid WGWO method showed better performance in terms of finding the optimal solution of the economic dispatch problem when compared with the other methods.

In Ref. [60], a novel power system planning strategy has been presented by integrating two algorithms of WOA and Pattern Search algorithm (PS). The WOA-PS has been done on the IEEE 30-bus test system with regards to several objective functions including generating fuel cost, voltage profile improvement, minimization of total power losses and emission reduction are also reviewed. The findings are compared with newly published meta-heuristic algorithms. Simulation findings totally demonstrate the efficacy and the speed of the proposed algorithm for

solving the Optimal Power Flow (OPF) problem.

In Ref. [61], a novel hybrid algorithm termed as Sine-Cosine WOA (SCWOA) is used for parameter optimization problem of multi-pass milling process so that to decrease total production time. Adding the updating operator of the SCA [62] instead of shrinking encircling mechanism of the WOA is the initial betterment. This assists the WOA to form an effective trade between both investigation and exploitation in introductory cycles in terms of which the spiral movements of whales can ensure the investigation of the arrangement space. The second betterment concerns WOA, and instead of this behavior of the humpback whales, in SCWOA the concept behind SCA has been utilized in order to force all the solutions to upgrade their position with regard to the best solution achieved so far. It is guaranteed that all the whales in the SCWOA upgraded their position regarding the best humpback whale. The SCWOA uses exploration and exploitation abilities of the two basic algorithms to gain better solutions. To represent the adequacy of the proposed algorithm, a treatment is done and the result of the proposed algorithm is compared to the other studies in the literature. The findings showed that SCWOA supplies promising solutions that this lead to minimum production time that is significant.

In Ref. [63], Brain Storm Optimization (BSO) [64] which is an optimization scheme based on WOA (BS-WOA), has been introduced to identify the secret key. The database from the data owner is modified with the optimal secret key for constructing the retrievable perturbation data for preserving the privacy and utility. The BS-WOA uses the fitness function for deriving the secret key, namely, the privacy and the utility of the data are kept as high as possible. We design the proposed BS-WOA through the hybridization of BSO and WOA. The proposed technique with the BS-WOA is simulated in the three standard databases, such as chess T10I4D100 K, and the retail databases. Upon evaluation for the key size of 256, the proposed BS-WOA achieved privacy value of 0.186 and utility value of 0.8777 for the chess database, hence enhancing the performance.

These methods are on the basis of probability and hold a population of candidates. Utilizing population sets helps the algorithm prevent becoming trapped at a local optimum. Furthermore, the hybrid method applies a local optimization method which starts from the best point in the population set in order to achieve faster convergence. Such hybrid algorithms keep a trade-off between accuracy and computational cost. Hybridization of WOA with Meta-Heuristic Algorithms is shown in Table 1.

In the vast majority of the papers that have been reviewed, WOA is implemented without cohesion with other meta-heuristic algorithms. For instance, PSO or GWO can be utilized as starting point to produce a good initial solution and rest of the search can be taken care by the WOA. In some cases, WOA can be used as a starting point to produce an initial solution and others can be used conduct the search. Moreover, the deterministic local search (LS) may be hybridized into the WOA as a supplementary tool to elevate the capability of exploitation and produce high quality solutions. Additionally, it may be useful to improve the search speed.

One of the most interesting trends in the last years is on hybrid algorithms. Indeed, an increasing number of papers are published about the hybridization of meta-heuristics with other models for optimization. Hybridization is not restricted to the combination of different meta-heuristics but includes the use of hybrid algorithms that combine local search or exact algorithms and meta-heuristics.

3.1.2. Artificial neural networks

The Artificial Neural Networks (ANNs) are competent of learning via training [65]. It is needed to train a set of data for this aim that are a series of input and output vectors. At training time, the ANNs is presented in iteration with the training data, and weights in the network are balanced from time to time so that the desired input-output mapping happens. After the training, the process will be as follows:

Input vectors that do not belong to the training pairs will be used to

Table 1
Hybridization of WOA with meta-heuristic algorithms.

Ref	Proposed Algorithm	Hybrid	Description	Application	Results
[48]	HWPSO	PSO	Two novel techniques have been employed, namely 'Forced' Whale and 'Capping' Phenomenon. The 'Forced' WOA is introduced in exploration phase and it enables the WOA to guide PSO for better local optima avoidance and concurrently a new 'Capping' phenomenon is employed for exploitation phase.	electronic design optimization problems	To improve the quality of the solution. The effectiveness of the proposed HWPSO has been also validated by statistical as well as complexity analysis.
[49]	HPSO-WOA	PSO	WOA is applied for exploration as it uses logarithmic spiral path so covers large uncertain search space with less computational time to explore possible solution or to converse particle toward optimum value.	Global Numerical Functions Optimization	optimum solution
[50]	HWA	Local Search	HWA can converge quickly toward optimal solutions using different techniques such as local search strategy, swap mutation, insert-reversed block operation and combining a heuristic algorithm called NEH.	WOA with a local search strategy for solving the permutation flow shop scheduling problem	Proposed method achieved better performance in comparison with other methods
[51]	HAGWO	GWO	WOA is used for the exploration phase as it uses a spiral function, hence, it covers a broader area in an uncertain search space	Global Optimization	superior quality of solutions with a minimum number of iterations
[52]	WOA-CBO	CBO	In order to modify and improve the updating mechanism of the standard WOA, a modified version of the WOA is considered	construction site layout planning problem	better solution with more stability
[54]	MWOA	Memetic	It is expected that MWOA can well balance the global exploration and local exploitation during the search process, thus achieving a better search performance.	Global Optimization	convergence speed and the solution accuracy
[55]	WOA-SA	SA	The proposed approach aims to enhance the exploitation of the WOA algorithm.	feature selection	WOA-SA in improving the classification accuracy compared to other wrapper-based algorithms
[56]	WOA-MFO	MFO	Optimization for multilevel thresholding image segmentation	Image segmentation	good balance between exploration and exploitation in all images at small and high threshold numbers
[58]	WGC	GWO	The WGC algorithm computes the optimal centroid for performing the data clustering	data clustering	The proposed WGC attained a maximum value of F-measure, rand coefficient, and Jaccard coefficient
[59]	IWOA-GWO	GWO	The proposed method efficiently integrates the mechanisms of WOA and GWO with crossover and mutation operators.	economic dispatch	better performance to find the optimal solution
[60]	WOA-PS	PS	Final values of the obtained parameters by WOA are regarded as the initial points for the algorithm of PS.	optimal power flow problem	the effectiveness and the rapidity of the proposed algorithm for solving the OPF problem
[61]	SCWOA	SCO	adding the updating operator of the SCA instead of shrinking encircling mechanism of the WOA	Parameter optimization	The proposed SCWOA performs significantly better than SCA and WOA
[63]	BS-WOA	BSO	The BS-WOA, is introduced for identifying the secret key. The BSWOA identifies the elements of the optimal key matrix.	Privacy preservation	better performance

simulate the system and create the corresponding output vector. Via choosing a suitable set of weight and transferring function, it was demonstrated that the ANNs can inexact any smooth, quantifiable function between the input and output vectors. The neural systems are broadly connected in numerous ranges like prediction, system modeling and control.

In this paper, an effective method is presented for purposes of automatic segmentation and classification of the brain MRI images by applying Modified Level Set approach and Adaptive ANN [66]. In the hidden layer, WOA is applied by the proposed Adaptive ANN for determining the optimal layer neuron groups. In addition, the Multi-level wavelet based features are applied to predict the image classes accurately. The performance of the proposed WOA-ANN is compared with those of GWO-ANN and ANN methods as regards sensitivity, specificity, and accuracy measures. The result outcome indicates that nearly 98% accuracy is obtained for the proposed method; however, it is 92% for GWO-ANN and 68% for default ANN.

In Ref. [67], a vertical handover model has been applied for the heterogeneous and diverse network with WIMAX and WIFI, with the assistance of the WOA-NN method. As a matter of fact, ANN has been utilized to achieve the knowledge in terms of the features of the network to distinguish the handover points. At first, the RSS of the heterogeneous network has been noticed and besides, the training library has been made. Consequently, RSS has been predicted by the trained network. Furthermore, WOA optimization has been used to make sure of the accuracy of learning of the neural network about the RSS characteristics of

the network. When simulated, the proposed WOA-NN model and its performance have been compared with the traditional models of LM-NN, FF-NN, PSO-NN, and GWO-NN. In the performance analysis, the throughput, handover predicted RSS and after that MAE have been analyzed. The MAE of the proposed WOA-NN model has been obtained to be 73.88% lesser than LM-NN, 36.91% lesser than FF-NN, 11.52% lesser than PSO-NN and 19.86% lesser than the GWO-NN models for the network configuration with WIMAX 5, WIFI 15 and users 30. Therefore, through observation it is found that RSS predicted by the proposed WOA-NN model is better than the conventional models, in providing powerful handoff.

In Ref. [68], combinations of WOA and back propagation ANN methodology are integrated with the aim of diagnosing diabetes mellitus. The mentioned proposed method underpins high convergence speed and improved precision. Based on this combination, local minima trapping problem which affects the quality of the solution is decreased in general. In the proposed methodology, new solutions are developed by WOA in solution space and the globally optimal solution has been found by back-propagation algorithm. The studies compare the proposed methodology with other algorithms and as a result, it is revealed by the findings that the performance of the proposed algorithm is higher than the other methodologies.

In Ref. [69], a new training algorithm in accordance with the recently proposed WOA has been suggested. It was revealed that this algorithm is capable of solving a vast range of optimization problems and its performance is higher than the current algorithms. The set of 20 datasets with

different levels of difficulty are chosen to test the proposed WOA-based trainer. While comparing back-propagation ANN algorithm with six evolutionary techniques, the findings are confirmed. Both qualitative and quantitative findings are a proof that the proposed trainer outperforms the current algorithms on the most of datasets with respect to both local optima avoidance and convergence speed.

Hybridization of WOA with ANNs is demonstrated in Table 2.

3.1.3. Support Vector Machine

Support Vector Machine (SVM) [70] is a supervised machine learning algorithm. It is designed to perform the two tasks of classification and regression, by analyzing the data and exploring specific hidden or visible patterns. In addition to the linear classification, SVM is known for its potential to perform tasks such as nonlinear classification tasks; because it works by dispersing the training datasets into another dimensional space, and performs linear separation on the data to investigate any existing categories. The separation process is done by making a hyper plane purported at increasing the margin between the nearest points of various categories known as support vectors to explore the best separating hyper plane.

In Ref. [71], a method has been proposed for medical data categorization using a novel and new ontology and WOA-based SVM (WOA-SVM) approach. At first, adopting Kronecker product (In mathematics, the Kronecker product, denoted by $\otimes$, is an operation on two matrices of arbitrary size resulting in a block matrix) bat algorithm, privacy-preserved data are advanced, and ontology is made for the sake of feature selection process. Afterwards, Ontology and WOA-based SVM are offered by combining ontology and WOA into SVM, where ontology and WOA is used for the possible selection of kernel parameters. The study is carried out using heart disease datasets, their numbers are 3 and they are datasets of Cleveland, Switzerland, and Hungarian. In a comparative study, the performance of the WOA-SVM approach is contrasted and compared with approaches of K-nearest neighbor, Naive Bayes, decision tree, SVM, and WOA-SVM. The evaluation metrics are accuracy, sensitivity, specificity, and fitness. The OW-SVM approach has the ability to achieve high performance with accuracy of 83.21%, the sensitivity of 91.49%, specificity of 73%, and fitness of 81.955, hence, it outperforms to the current comparative techniques.

Table 2
Hybridization of WOA with ANNs.

Ref	Proposed Algorithm	Description	Application	Results
[66]	WOA + ANN	In the ANN, the layer neurons are optimized using WOA	image segmentation	increase the classification accuracy
[67]	WOA + NN	By exploiting WOA, the error performance of learning algorithm is measured, and the updated weight may have minimized error, which is given for training.	Heterogeneous network	attaining effective handoff
[68]	WOA + ANN	The WOA develops new solutions in solution space and backpropagation algorithm finds the globally optimal solution	Classification	high convergence speed and improved accuracy
[69]	WOA + ANN	The WOA was employed to find the best values for weights and biases to minimize the MSE	Optimization	local optima avoidance and convergence speed

In Ref. [72], a novel machine learning model in terms of SVM and a new metaheuristic algorithm known as WOA has been used so that to identify spammers on Twitter. The SVM-WOA is utilized to reveal the foremost compelling features in the process of recognizing the spammers. Two simultaneous tasks are the tasks that are used in this model. The first task tries to optimize the parameters of the SVM, while the second task acts as the feature selection task. The proposed approach was used on diverse lingual contexts such as Arabic, Spanish, English, and Korean. The results of studies revealed that the model outperforms significantly in comparison with many other classical models. The models that prevalently used in the literature for the sake of diagnosing spammers in online social networks.

A computational model has been proposed based on WOA + SVM [73] for predicting the toxicity effects of Bio transformed Hepatic Drugs. The proposed model includes three phases. In the first stage, the highly discriminative subset of features is chosen through rough set-based methods to decrease the categorization time while it improves the classification performance. In the second stage, various sampling methods are utilized to solve imbalanced dataset issues. The sampling methods were as follows:

Random Under-Sampling, Random Over-Sampling and Synthetic Minority Oversampling Technique (SMOTE), Borderline SMOTE and Safe Level SMOTE.

In the third stage or phase, the SVM classifier is used so that to divide an unknown drug into toxic or non-toxic. Two SVM parameters that are the penalty parameter and kernel parameter influence the classification accuracy of the model. In this paper, WOA has been used for optimizing the parameters of SVM in order to decrease the classification error. The results of studies approved that the proposed model achieved high sensitivity to all toxic effects. On the whole, the high sensitivity of the WOA + SVM model shows that this hybrid model could be used for the tasks such as prediction of drug toxicity in the early phase of drug development.

In Ref. [74] a model has been proposed for categorization, exploiting the SVM principles in order to find optimized structures of the Enron-Spam dataset that have high similarity. The WOA is fostered to gain an optimal feature subset for E-mail classification. In the study, four various kernel functions are exploited. The functions contain Linear, Quadratic, Polynomial, and RBF in classification so that to test the best kernel function for SVM. Different and various evaluation measurements are calculated in order to explore the performance of the proposed technique. From the evaluation measurements, we can name Precision, Accuracy, Recall, and F-measure. The results of experiments are analyzed, contrasted and compared with those from other techniques published in spam E-mails filtering. The results of comparisons and analysis showed that WOA + SVM are very useful and competitive for E-mail classification.

Hybridization of WOA with SVM is demonstrated in Table 3.

With a good performance and flexibility, WOA is an algorithm that can be applied to solve different types of problems from many areas. Meta-heuristic, ANNs and SVM have been compared. Comparison of the Meta-heuristic and other method indicate that the Meta-heuristic is more positive. Percentage of WOA hybrid papers with meta-heuristic, ANNs, and SVM is indicated in Fig. 8. According to the statistical result shown in Fig. 8, the meta-heuristic algorithms have more percentage.

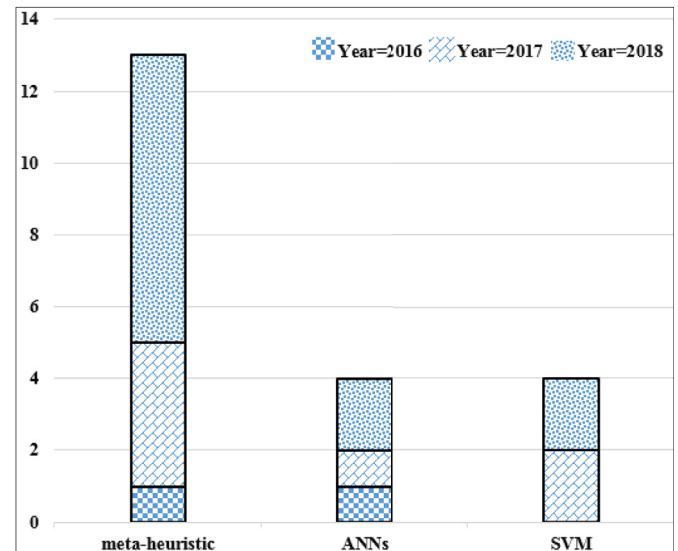
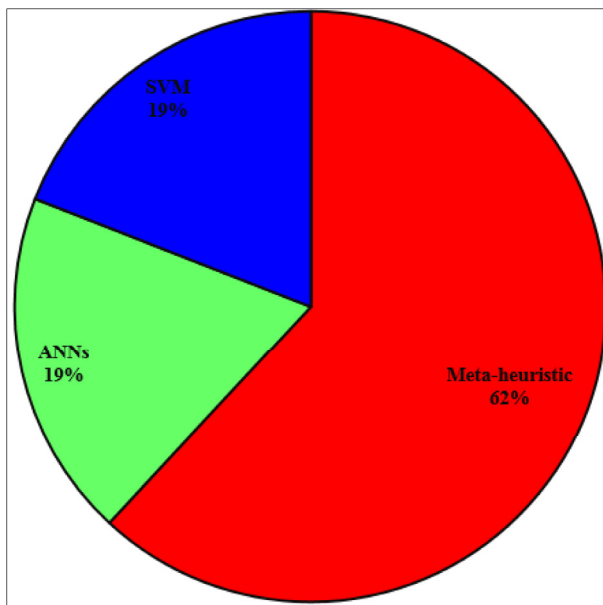
Fig. (9) shows the Flowchart of the number of hybrid WOA papers based on publication year. As can be seen in the flowchart, it is clear that maximum number of articles published in 2018 belong to the field of metaheuristic algorithms. The maximum hybridization with WOA is related to metaheuristic algorithms.

3.2. Improved WOA

This section is related to improving WOA using Chaotic, Levy Flight, and Opposition-based Learning.

Table 3
Hybridization of WOA with SVM.

Ref	Proposed Algorithm	Description	Application	Results
[71]	WOA-SVM	The WOA is used for the feasible selection of kernel parameters	data classification	maximum performance
[72]	WOA + SVM	The WOA is utilized to optimize the parameters of SVM and search for the best subset of features.	spam profiles detection on online social networks	remarkably high accuracy percentage
[73]	WOA + SVM	In this paper, WOA has been proposed to optimize the parameters of SVM, so that the classification error can be reduced.	Classification	classification accuracy
[74]	WOA + SVM	The WOA was used for obtain an optimal feature subset for E-mail classification	E-mail Classification	increase performance

**Fig. 9.** Flowchart of the number of WOA papers based on publication year.**Fig. 8.** Percentage of WOA hybrid papers with meta-heuristic, ANNs, and SVM.

3.2.1. Chaotic

Despite the fact that WOA has suitable convergence rate, WOA still cannot perform as effectively as possible when it comes to find the global optima which influence the convergence rate of the algorithm. So, to reduce this affect and to improve its capability and efficiency, Chaotic WOA (CWOA) [86] algorithm is developed by introducing chaos in WOA algorithm itself. In general, chaotic rooted from the term ‘chaos’ defined as the property of a complicated system with unpredictable behavior, and map is defines as mapping or associating chaos behavior in the algorithm with some parameter using a function. Because of the periodicity and non-repetition properties of chaos, it can perform general searches at

higher speeds when it compared to the stochastic searches that basically depend on probabilities [86]. Chaotic maps are the maps that represent complex and dynamic behavior in the non-linear systems.

The meaning of chaos is that the dynamic behavior that is not stable provides sensitive dependence on the first conditions and includes unlimited unstable periodic motions in the systems that are nonlinear. The three critical dynamic characteristics of chaos with fine internal structure are periodicity, the sensitive reliance on initial conditions, and the quasi-stochastic property. The first property is the critical one and all nodes or states in the search plane can be searched by this property by its formulas within specific range. Chaos strategy is applied to prevent being trapped in local optima and the quality of searching global optimum can be improved by this strategy. Hence, chaos has been used in various optimization applications. The optimization procedure of the CWOA is also shown using the following flowchart that has been presented in Fig. 10 [86].

Chaotic maps utilizing their dynamic behavior have been vastly admitted and acknowledged in the area of optimization so that to assist optimization algorithm in investigating the search space more energetically and globally [86]. In nearly all meta-heuristic algorithms containing stochastic components, randomness is obtained by applying probability distributions. It would have benefits to replace such randomness by utilizing chaotic maps. For the aim of introducing chaos in optimization algorithm, various chaotic maps with different mathematical formulas or equations are used as the following list in Table 4.

Since previous years, chaotic maps have been vastly well received in the area of optimization algorithms in investigating the search space more vigorously and globally. Commensurate with various human's domain a large variety of chaotic maps prepared by physicians, researchers, and mathematicians are accessible in the optimization area. From among these, the most important one-dimensional chaotic maps that their numbers are ten [86] have been used in the present study to trigger CWOA, details of which are given in Table 1. The convergence rate of WOA has been significantly and positively affected by applying chaotic maps because these maps persuade chaos in the possible area as predicted only for not long and initial time and is stochastic for a longer period of time [86]. Pseudo code of the proposed CWOA algorithm is represented in Algorithm 2.

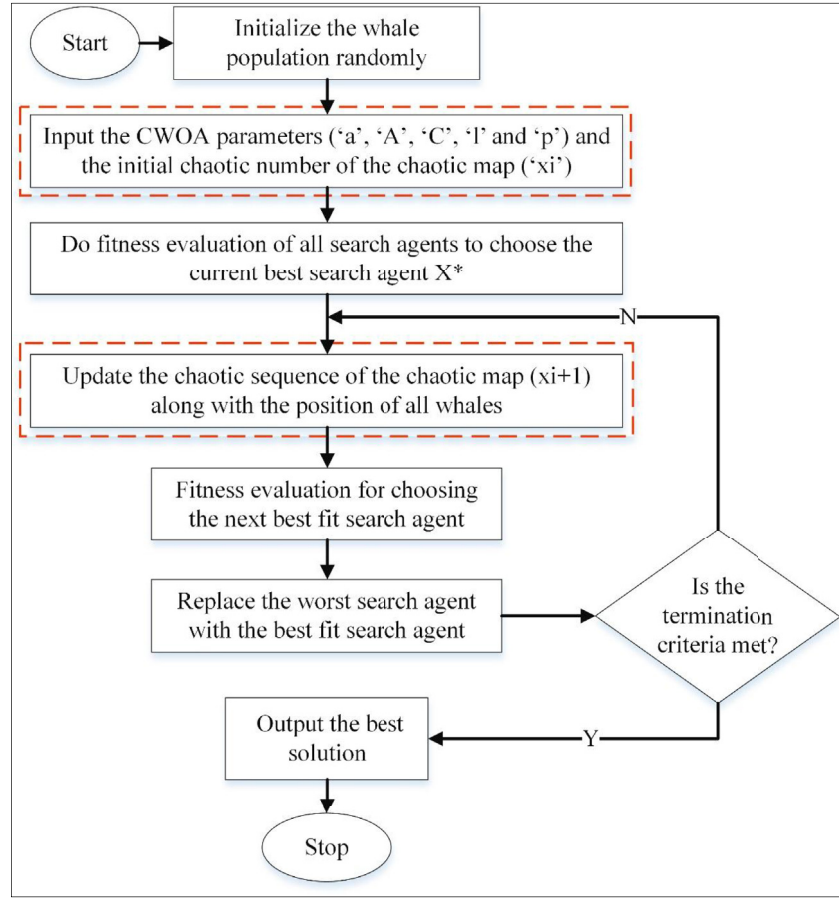


Fig. 10. Flowchart of optimization procedure of CWOA [86].

Algorithm 2

Pseudo-code of CWOA algorithm [86].

Initialize the generation counter t and randomly initialize the whale's population $X_i(i = 1, 2, \dots, n)$

Evaluate the fitness of each search agent to find the best search agent X^*

Initialize the value of the chaotic map x_0 randomly

while ($t < \text{maximum number of iterations}$)

Update the chaotic number using the respective **chaotic map** equation

for each search agent

Update a , A , C , l and p

if1 ($p < 0.5$)

if2 ($|A| < 1$)

Update the position of the current search agent by Eq. (1)

else if2 ($|A| \geq 1$)

Select a random search agent (X_{rand})

Update the position of the current search agent by Eq. (8)

end if2

else if1 ($p \geq 0.5$)

Update the position of the current search by Eq. (5)

end if1

end for

Check if any search agent goes beyond the search space and amend it

Calculate the fitness of each search agent

Update X^* if there is a better solution

$t = t + 1$

end while

return X^*

In Ref. [86], with the aim of designing an improved CWOA chaos theory and WOA has been hybridized. For adjusting the key parameter, p , of WOA, very different and vast chaotic maps have been used. In order to compare and verify the performance of CWOAs, the benchmark functions that their numbers reach to twenty, subcategorize into multimodal and

Table 4

Chaotic maps [86].

No.	Name	Chaotic Map
1	Logistic	$X_{i+1} = ax_i(1 - x_i)$
2	Cubic	$X_{i+1} = ax_i(1 - x_i^2)$
3	Sine	$X_{i+1} = \frac{a}{4} \sin(\pi x_i)$
4	Sinusoidal	$X_{i+1} = ax_i^2 \sin(\pi x_i)$
5	Singer	$X_{i+1} = \mu(7.86x_i - 23.31x_i^2 + 28.75x_i^3 - 13.302875x_i^4), \mu = 1.07$
6	Circle	$X_{i+1} = \text{mod}\left(x_i + b - \left(\frac{a}{2\pi}\right) \sin(2\pi x_k), 1\right)$
7	Iterative	$X_{i+1} = \sin\left(\frac{a\pi}{x_i}\right)$
8	Tent	$X_i = \begin{cases} x_i/0.7 & x_i < 0.7 \\ \frac{10}{3}(1 - x_i) & x_i \geq 0.7 \end{cases}$

unimodal problems. In general, the results revealed that chaotic maps are capable of significantly improving the performance of WOA. Among all the chaotic maps, the Tent map has significantly increases the performance of WOA. The foremost reason of the higher performance of CWOA is the chaos contained by the chaotic maps in the search space. The chaos assists the controlling parameter with searching the optimal solution in a fast manner and refining the convergence rate of the algorithm.

In Ref. [87], a hybrid method has been proposed to hybridize the WOA and chaotic optimization. So far the WOA algorithm has not emphasized in the literature related to multiple input and multiple output (MIMO) radar rate. In order to introduce chaotic optimization into the WOA scheme via their simulation, it is represented that the proposed algorithm is possible and affective for array synthesis of MIMO radar. In

detail, the proposed hybrid method can act as a block for the premature of origin algorithm and apply desired side lobe suppression with the limitations of broad nulls (BNs). Regardless of other existing algorithms used in MIMO radar, the proposed algorithm only takes <100 steps of repetition to find the better solution.

In Ref. [88], a new CWOA has been applied to tackle these problems in the case that chaotic search is embedded in the searching repetitions of WOA. Ten chaotic maps are regarded to enhance the accuracy of WOA. Experimental studies on ten benchmark datasets show the novel CWOA is more influential in selecting detailed and accurate features with a high sub categorization performance and a small number of features. Furthermore, the performance of CWOA is contrasted and compared with WOA and the other ten optimization algorithms. The experimental findings are representative of the fact that circle chaotic map can be considered as the best chaotic map so that to significantly increase the grade of the WOA performance. What is more is chaotic with modifications of investigation operators outperforming the performance to the higher degree.

In Ref. [89], the CWOA for the parameters assessment of solar cells has been proposed. The essential merit of the proposed approach is utilizing the chaotic maps in order to commute and automatically adapt the internal parameters of the optimization algorithm. This situation is influential in complicated problems, since along the iterative process, the proposed algorithm increases their abilities to search for the best solution. The modified method is capable of optimizing complex and multimodal objective functions, for example, the function for the measurement of parameters of solar cells. The proposed algorithm in the solar cell design is contrasted with other optimization methods over various datasets to imagine the abilities of this algorithm. In addition, the experimental studies advocate the performed performance of the proposed approach in terms of accuracy and robustness.

In Ref. [90], a combination of the chaotic concept with basic WOA has been made available to add some positive aspect, i.e. enhanced computational speed, improved convergence profile while applying in Transient stability constrained OPF (TSCOPF) problem of power systems. The adaptation of CWOA is conducted in this paper to exercise on Western Systems Coordinating Council (WSCC) 3-generator, 9-bus, and IEEE 30-bus test power systems for the aim of TSCOPF study. The simulation findings obtained drawn from the proposed CWOA strategies, are compared to those offered by some other computational intelligence based techniques rooted in the recent state-of-the-art literature containing the basic WOA one.

In Ref. [91], chaos-based WOA has been used for TSCOPF problem of power system considering multiple contingencies. The proposed algorithm is assessed on two various test power systems. Simulation findings, as yielded by the proposed CWOA, are compared to other current algorithms rooted from the recent literature. Basic WOA-based findings are also explored and contained in this research. From the findings, it is approved that, with the existence of chaotic variables in the basic WOA method and the performance in the convergence profile has been presented to and the transient performance of the power system is improved as a results of harmonious and balanced trade-off or interaction between exploration and exploitation.

In Ref. [92], a CWOA-Elman soft-sensor model has been utilized to prognosticate the conversion velocity of vinyl chloride monomer (VCM). Elman ANN with powerful nonlinear performance is selected for the construction of the soft-sensor model. Nonetheless, because of the early phase of Elman neural network to train or educate the connection weights of the layers, the training effect is cumbersome to be guaranteed or ensured with the connection weights. Hence, the WOA is adopted to optimize Elman ANN with the purpose of avoiding falling into the local optimum. Regarding the good nonlinear approximation ability of Elman ANN as well as the simplicity and easiness to understand and gain the global extreme, avoid local extreme and have the certain characteristics of the adaptive ability to the search space of WOA, CWOA is proposed to optimize the connection weights of Elman ANN among layers. The

Table 5
Improvement of WOA with chaotic.

Ref	Proposed Algorithm	Description	Application	Results
[86]	CWOA	chaotic helps in controlling exploration and exploitation	Optimization Problems	Increase efficiency
[87]	CWOA	By introducing chaotic optimization into the WOA scheme, performance of origin algorithm is improved substantially in suppressing peak side lobe level with constraint of broad nulls.	Side lobe suppression with constraint for MIMO radar	good performance
[88]	CWOA	Chaotic is employed to manipulate the random parameters values of WOA. Chaotic with modifications of exploration operators outperform the highest performance.	Features Selection	improve significantly the performance of the WOA
[89]	CWOA	The main advantage of the proposed approach is using the chaotic maps to compute and automatically adapt the internal parameters of WOA	Parameter estimation of photovoltaic cells	Improved performance of the proposed approach regarding accuracy and robustness.
[90]	CWOA	Chaotic is employed to manipulate the random parameters values of WOA.	optimal power flow	enhanced computational speed, improved convergence
[91]	CWOA	Chaotic is employed to manipulate the random parameters values of WOA.	transient stability constrained optimal power flow	improvement in the convergence
[92]	CWOA	CWOA) based the idea of chaos is proposed in order to improve the diversity of all search agents and egocentricity of agent search by utilizing the chaotic features.	polymerization process	prediction accuracy

simulation findings have revealed that the soft-sensor model according to the proposed CWOA-Elman ANN represented high prediction accuracy.

Improvement of WOA with chaotic is shown in Table 5.

3.2.2. Levy flight

Levy flight is a specific class of random walk that the step lengths are distributed based on a heavy power law tails. The large steps occasionally assist an algorithm to do or conduct a global search. It would be useful to apply the Levy flight trajectory [98,99] in gaining a better trade-off between the exploration and exploitation of the algorithm, and it has plus

points based on the avoidance of local optimal optima.

In Ref. [100], performance has been proposed to the WOA that is according to a Levy flight trajectory and named the Levy flight trajectory-based WOA (LWOA). The LWOA makes the movement of WOA faster and more powerful and prevents premature convergence. The Levy flight trajectory can be utilized to enhance the diversity of the population in contrast to premature convergence and performs the ability of jumping out of local optimal optima. This method assists the exploration and exploitation of the WOA in order to obtain a better tradeoff between them. The LWOA is characterized by fast convergence and high accuracy or precision, and it can effectively go away from or get rid of a local optimum. The LWOA is also compared with other well-known nature-inspired algorithms on 23 criteria. The reason is to solve infinite impulse response model identification. The results of statistical data analysis on the benchmark functions represented that the LWOA can significantly increase the performance of others on a majority of the benchmark functions, especially in resolving an optimization problem that this problem or issue has high dimensionality. Furthermore, the superior identification capability of the LWOA is clear from the results obtained through the simulation study in comparison with other algorithms. All the results are indicative of the superiority of the LWOA.

A modified WOA (MWOA) has been proposed in order to solve large-scale global optimization (LSGO) problems [101]. In order to create balance between the exploration and exploitation abilities, a nonlinear dynamic strategy is proposed or utilized regarding a cosine function for updating the control parameter. A Levy-flight strategy is adopted to make the algorithm jump out of local optima. Furthermore, a quadratic interpolation method is used to the leader of the population in order to improve the local exploitation ability as well as to upgrade the solution accuracy. MWOA is examined over 25 well-known benchmark functions with dimensions ranging from 100 to 1000. The results of experimental studies show that the superior performance of MWOA is compared with other state-of-the-art optimization algorithms on LSGO, with respect to solution accuracy, convergence speed, and stability.

In Ref. [102], it has been focused on solving virtual machine (VM) placement problem in terms of the available bandwidth formulated as variable sized bin packing problem. Besides, a new bandwidth allocation policy is developed and hybridized with an improved variant of WOA called improved Levy based WOA. The validity of the proposed algorithm Cloudsim toolkit has been tested on 25 different data sets that generated randomly and compared with many optimization algorithms including: WOA, first fit, best fit, PSO, GA, and intelligent tuned HSA. The obtained results are analyzed by Friedman test showing the prosperity of the proposed algorithm to minimize the number of running physical machine.

In Ref. [103], an improved WOA (IWOA) has been proposed to solve both single and multidimensional 0–1 knapsack problems with different scales. A penalty function is added to the evaluation function so that the fitness of the feasible solutions can outperform the fitness of the infeasible ones. The real-valued solutions can be taken by sigmoid function as input and binary solutions can be produced as output. A two-stage repair operator is employed to handle the infeasible solutions. Local Search Strategy (LSS) and the Levy flight walks are the two strategies through which IWOA can give a better tradeoff between the diversification and the intensification. In addition the bitwise operation increases the IWOA efficiency. The proposed IWOA is compared with other state-of-art algorithms to validate the efficacy of the proposed algorithm in solving 0–1 knapsack problems. The experimental results and extensive numerical illustrations have demonstrated that IWOA is efficient, effective, and robust in order to solve the hard 0–1 knapsack problems than the other existing approaches in the available literature.

Improvement of WOA with levy flight is illustrated in Table 6.

3.2.3. Opposition-based learning

Tizhoosh originally introduced the basic concept of OBL in Ref. [115]. Fitness of the current individual and the opposite individual is calculated

Table 6
Improvement of WOA with levy flight.

Ref	Proposed Algorithm	Description	Application	Results
[100]	LWOA	The Levy flight trajectory is helpful for increasing the diversity of the population against premature convergence and enhancing the capability of jumping out of local optimal optima.	Global Optimization	superior performance
[101]	MWOA	Levy-flight strategy is adopted to make the algorithm jump out of local optima.	large-scale global optimization problems	solution accuracy, speed convergence, and stability
[102]	ILWOA	The discovery of candidate solutions needs a more random way of moving through the search space. So, the whale movement is adjusted with a more diffusion distribution called Levy flight.	bandwidth-efficient virtual machine placement in cloud computing environment	superior performance: Minimizing the number of running physical machine
[103]	IWOA	The Levy flight is added to the IWOA to accelerate the convergence of IWOA toward the optimal solution. As well as, it can escape from the local optimum solutions because of its ability to explore new regions in the search space for finding better solutions.	knapsack problem	best solution

by the basic idea of OBL [197]. Then, the individual with the lower (higher) fitness value is selected by the algorithm.

In Ref. [116], an improved version of the WOA has been proposed using the opposition-based learning (OBL) to better the exploration of the search space. Three different diode models are used to estimate the parameters of solar cells with this algorithm. Such models are the single diode model, the double diode model and the three diode model, each of which having different internal parameters that has to be accurately estimated in order to have a good performance of the solar cells. The inclusion of the three diode model is due; it offers a more accurate representation of the solar cells behavior in industrial applications. Similar approaches and datasets are used from solar cells and photovoltaic modules for experiments and comparisons. Moreover, the proposed method has also been examined over different benchmark optimization functions for the verification of its exploration capabilities. The experiments and comparisons are supportive of the performance of the proposed approach in terms of complex optimization problems.

The population of WOA is randomly generated during the exploration and exploitation phases, which could produce values far from the

optimum solution or stuck the exploration around local optima. A new algorithm has been proposed based on WOA in order to enhance solution accuracy and reliability, [117]. The new algorithm is known as Opposition-based WOA (OWOA). The OBL method is used by the OWOA to for betterment of WOA performance. Search for the solution is carried out by the OWOA in the opposite direction of suggested values to test if the opposite choice has better solution. The OWOA is tested and compared with the original algorithm WOA and other metaheuristic methods. The efficiency of the OWOA which is more efficient is confirmed by the benchmark results.

According to the statistical result shown in Fig. 11, the chaotic methods have more percentage on WOA improvement papers.

3.3. Variants of WOA

This section describes different variants of the WOA algorithm.

3.3.1. Binary

A variety of optimization problems cannot be handled by continuous meta-heuristics. A scientific field called Binary optimization is in combinatorial optimization. Binary optimization problems have distinct decision variables that are the elements of set [0, 1]. In short, each decision variable of the binary optimization problem can be either 0 or 1. 0 or 1 can be considered as a digital meaning of the decision variables in terms of the binary optimization problem. Problems such as Feature Selection (FS) [75,76], unit commitment problem [77,78], and knapsack problems (KP) [79,80], experienced in an expansive number of cases, are modeled as a binary optimization problem in reality.

In Ref. [81], a novel and new binary version of WOA (BWOA) has been used to choose the optimal feature subset for dimensionality reduction and classifications problem. The novel approach is accordance to a sigmoid transfer function (S-shape). By managing with the FS problem, a free position of the whale must be transformed or changed to their corresponding binary solutions. This transformation is performed by using an S-shaped transfer function in each dimension that characterizes the likelihood of changing the position vectors' elements from 0 to 1 and vice versa and consequently oblige the search agents to move in a binary space. K-NN classifier is used to guarantee that the chosen highlights are the pertinent ones. A set of criteria are used to assess and compare the proposed BWOA-S with the native one over eleven different datasets. The results confirmed that the new algorithm has a significant performance in finding the optimal feature.

In Ref. [82], the binary WOA (BWOA) has been proposed, to solve intricate, limited, non-convex, binary-nature profit-based unit commitment (PBUC) optimization problems of a price-taking generation company in the electricity market. To simulate the binary-nature PBUC problem, the continuous, real-value whale position/location is mapped into binary search space through various transfer functions.

This paper presents three variables of BWOA using tangential hyperbolic, inverse tangent (arc tan) and sigmoidal transfer functions. The adequacy of the BWOA approaches is explored in test systems with various market mechanisms that are an energy-only market, and energy and reserve market participation with different and various reserve payment methods. The simulation findings are presented, discussed, contrasted, and compared with other existing approaches. The merging characteristics, arrangement quality, and consistency of the results over diverse BWOA variations are talked about.

In Ref. [83], a novel binary version of the WOA has been used to select the optimal feature subset and increase the accuracy of classification. The execution of the proposed binary WOA (BWO) is confirmed by comparisons with three famous optimization based FS algorithms; GA, ACO and PSO; on nine benchmark datasets. The results of both qualitative and quantitative data analysis demonstrate the potential ability of the proposed BWO to search the feature space for optimal feature combinations. Furthermore, the findings show and justified that the proposed BWO has the capability to outperform the current algorithms on the most of datasets in accordance with factors such as average classification accuracy and convergence speed.

In Ref. [84], for solving FS problem, a new binary version of the basic WOA named BWOA-V has been proposed. V-shaped transfer functions are used to convert the local version of WOA to a binary version. To investigate the performance of BWOA-V, the experiments are carried out on 11 benchmark datasets from UCI datasets. In addition, five evaluation criteria and the Wilcoxon's rank-sum nonparametric statistical test were employed to evaluate different aspects of the contrasted and compared algorithms. The experiments results proved that the proposed binary algorithm has the ability to minimize the number of selected features and to maximize the classification accuracy in FS domain.

In Ref. [85], FSWOA has been proposed for FS. The performance of proposed algorithm is appraised and conducted by four standard medical datasets: Pima Indians Diabetes, Original Wisconsin Breast Cancer, Statlog and Hepatitis. The results demonstrate that the FSWOA can reduce the dimensionality of medical datasets with acceptable accuracy for diseases diagnosis.

Variants of WOA with binary are shown in Table 7.

3.3.2. Fuzzy logic

In Ref. [93], an adaptive fuzzy logic PID (AFPID) controller has been proposed with the aim of loading frequency control. A modified WOA (MWOA) was also utilized in this study for tuning of the AFPID parameters so that an enhanced performance might be achieved. The proposed algorithm, at the first stage, assessed using standard test functions and it was compared with other recent algorithms with the aim of authentication of algorithm competence. The proposed MWOA algorithm outperforms the algorithms of PSO, GSA, DE and FEP in more than 70% or five out of seven unimodal test functions and more than 60% or four out of six multimodal test functions. The efficacy of the AFPID was compared with the conventional PID and the results show the proposed AFPID had better performance. Reduction of 39.13% in error criteria (objective function) was compared with WOA-PID controller. Furthermore, the proposed approach was also compared with more recently proposed frequency control approaches in a vastly used two-area test system.

In Ref. [94], a dietary assessment system based on Cauchy, Generalized T-Student, and Wavelet kernel based Wu-and-Li Index Fuzzy clustering (CSW-WLIFC) based segmentation has been used and the proposed classifier is Whale Levenberg Marquardt Neural Network (WLM-NN). The proposed CSW-WLIFC based segmentation segments the image based on the current WLI-FC algorithm. A novel CSW based kernel function is

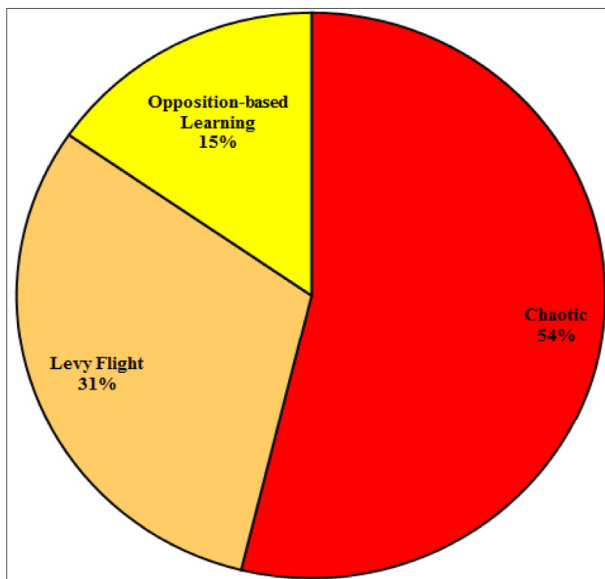


Fig. 11. Percentage of WOA improvement papers.

Table 7

Variants of WOA with binary.

Ref	Proposed Algorithm	Description	Convert	Application	Results
[81]	BWOA	The BWOA has been proposed to select the optimal feature subset for dimensionality reduction and classifications problem.	The conversion is performed by applying an S-shaped (sigmoid) transfer function.	Feature Selection	significant performance in finding the optimal feature
[82]	BWOA1 BWOA2 BWOA3	The variables and search space in this problem essentially consist of binary strings containing 0 and 1 bits	Tangent hyperbolic, inverse tangent (arc tan) and sigmoid transfer functions.	profit-based unit commitment problems	solution quality
[83]	BWOA	for selecting the optimal feature subset and increasing the classification accuracy	The conversion is performed by applying an S-shaped (sigmoid) transfer function.	Feature Selection	classification accuracy and convergence speed
[84]	BWOA	The BWOA has been proposed to select the optimal feature subset for dimensionality reduction and classifications problem.	Hyperbolic tangent function (V-shaped function)	Feature Selection	Classification accuracy within an appropriate time.
[85]	FSWOA	The FSWOA searches the problem space and extract a subset of optimal features for medical decision-making.	The conversion is performed by applying an S-shaped (sigmoid) transfer function.	Feature Selection	acceptable accuracy

utilized in the segmentation process. Feature vectors like color, form or shape, and texture are rooted from the segmented image. The ANN is trained with the Whale-Levenberg Marquardt (WLM) model to recognize each food item from the tray image. The calorie value of each food item is estimated by the proposed calorie estimator. Based on the simulation findings, it is clear that the proposed model has the increased performance than the existing models with the values of 0.999, 0.9643, 0.9627, and 0.0184 for the segmentation accuracy, macro average accuracy, standard accuracy, and mean square error, respectively.

In Ref. [95], a hybrid algorithm has been proposed through the concept of fuzziness and a novel P-Whale algorithm named Fuzzy Whale Fusion (FW-Fusion), for the fusion of MRI multimodal photos or images. Two multimodal images from MRI (T1, T1C, T2, and FLAIR) are thought to be the source images. They are fed as inputs to a wavelet transform. The images or photos are changed into four varied bands by the transform utilized. These images are fused and connected by two newly derived fusion factors, fuzzy fusion and whale fusion, in a weighted function. WOA and PSO are combined and connected by the proposed P-Whale approach for the efficient choice of whale fusion factors. The performance of FW-Fusion model is contrasted and compared to those of the existing strategies applying Mutual Information (MI), Peak Signal-to-Noise Ratio (PSNR) and Root Mean Squared Error (RMSE), in fact, these models were as the evaluation metrics. The results of observation from the mean performance evaluation show that the proposed approach can achieve MI of 1.714, RMSE of 1.9 and PSNR of 27.9472.

Modified kernel fuzzy c means (MKFCM) is used for test case prioritization [96]. The objective of test case prioritization is to determine test case ordering that maximizes the likelihood of discovering flaws in source code early. MANN classification algorithms are used to test case prioritization. In this case, WOA is applied for weight optimization process. As the result, score value is gained from the WOA based on the obtained score value, test case prioritization happens. So, our proposed method prioritizes the test cases efficiently and accurately. The main benefit of this method is that it is highly cost effective. The performance of the proposed method is estimated according to execution time. Furthermore, memory and Average Percentage of Faults Detected (APFD) are the other measurement tools.

In Ref. [97], an efficient approach has been proposed for binarization of handwritten Arabic manuscripts employing a WOA, incorporating a Fuzzy C-Means objective function, to obtain optimal thresholds. Experimental results prove the efficiency of the proposed approach in comparison with earlier methods.

Variants of WOA with fuzzy is demonstrated in Table 8.

3.3.3. Multi-objective

This section aims to introduce the basic concept of multi-objective problems (MOPs) [104]. The MOPs are aiming to minimize or

Table 8

Variants of WOA with fuzzy.

Ref	Proposed Algorithm	Description	Application	Results
[93]	MWOA: AFPID	WOA was proposed in this paper for tuning of the AFPID parameters.	fuzzy logic PID controller for load frequency control of autonomous power generation systems	better performance
[94]	CSW-WLIFC	The proposed Whale Levenberg Marquardt Neural Network (WLM-NN) classifier recognizes each food item with the use of the extracted features.	Fuzzy clustering and Whale-based neural network to food recognition and calorie estimation for daily dietary assessment	segmentation accuracy
[95]	FW-Fusion	A weighted function that combines fuzzy and whale fusion factors is used to obtain the wavelet coefficients.	Fuzzy Whale Fusion model for MRI multimodal image fusion	better performance
[96]	KFCM	Then the generated test cases are clustered with the aid of kernel fuzzy c-means clustering technique. WOA is used for weight optimization process.	test case prioritization using modified	very good accuracy
[97]	WOA-FCM	fitness function that is based on fuzzy c-means clustering algorithm	handwritten manuscript image binarization approach based on WOA and Fuzzy	better performance

maximize conflicting multiple objective functions [195,196]. Unlike single objective optimization problems, MOPs involve multiple objective functions that contrast each other. It is impossible to find a solution optimizing objective functions simultaneously. Hence, balance should be made between the objective functions to obtain an optimal solution set. It is required that a multi-objective optimization problem should have at least two objectives and some constraints. It is found at finding mutually agreeable solutions satisfying objectives and constraints in a solution space. To define the MOPs, consider the minimization problem with multiple functions $f_i(x)$, $i = 1, 2, \dots, M$ (where M is the number of

functions) as in Eq. (9) [104].

$$\text{Min } F(x) = [f_1(x), f_2(x), \dots, f_M(x)] \quad (9)$$

$$g_i(x) \leq 0, i = 1, 2, \dots, k \quad (10)$$

$$h_i(x) = 0, i = 1, 2, \dots, k \quad (11)$$

In Eqs. (10) and (11), x stands for the vector of solutions (decision variables), $g_i(x)$ and $h_i(x)$ represent the constraint functions. The solution x of Eq. (9) is said to dominate the solution y , if the conditions in Eq. (12) are satisfied. In Eq. (12) $i, j \in 1, 2, 3, \dots, M$.

$$\forall i : f_i(x) \leq f_i(y) \text{ and } \exists j : f_j(x) < f_j(y) \quad (12)$$

During the previous few decades, a wide range of multi-objective techniques have been developed. Multi-objective metaheuristic based techniques deal with many difficulties such as infeasible solutions, diversity of solutions, and optimum separation [105]. Two conditions require to be satisfied to involve with these problems or difficulties. First, the information is exchanged between the search space and search agents [106]. Second, the multi-objective techniques should assist to estimate the whole true Pareto optimal front in a single simulation run.

The common multi-objective metaheuristic is Non-dominated Sorting GA2 (NSGA-2) [107]. This algorithm utilizes a quick non-dominated sorting method. It is a technique that is more elite, and it is a niching operator. The population of NSGA-2 algorithm begins with randomization where each person is grouped based on the non-dominated sorting technique. Mutation and recombination an operator, the second population is created to help with the selection. Both of these populations create a novel one and so are re-sorted by a sorting technique that is non-dominated. The probability to select a new person relies on the non-domination level for the last or final population. At last, the all process is run so that the satisfactory result is obtained.

As the second common multi-objective metaheuristic technique, we can mention to Multi-objective PSO (MOPSO) [108]. It applies the concept of PSO algorithm. The external archive is used by MOPSO in order to store and retrieve the Pareto optimal solutions. In addition, to increase the randomness and diversity in the distribution of Pareto optimal solutions a mutation operator is compounded or integrated in MOPSO.

The other popular and common multi-objective metaheuristics is Multi-objective Evolutionary Algorithm according to Decomposition (MOEA/D) [109]. It decomposes a problem into scalar sub-problems so that to optimize at the same time or simultaneously. These sub-problems are usually equal to the population size at the place that each sub-problem is assigned a weighting vector to integrate the objective functions into a single objective function. Two tasks that are co-operation and completion are performed during search process. In co-operation task, the neighboring members can collaborate the sub-problem to generate solution. In addition, the solution of neighboring sub-problem(s) is replaced with the sub-problem solution in the case that it is better than the present one. This task is named competition task. MOEA/D has less computational complexity and fast convergence speed than NSGA-2.

The increased version of ACO Angus and Woodward [110] is used to solve multi-objective optimization problems. A new classification technique is used in Multi-objective Ant Colony Optimization (MOACO) that provides nearby solutions for intended problem domains. MOACO is based on the fundamental concepts of ACO. These are selection of pheromone model, construction process, solution estimation, and updating process. This technique has the ability to balance the exploratory (diversity preserving) behavior with exploitative (quality enhancing) behavior from an optimization viewpoint.

In Ref. [111], a new non-dominated sorting based on MOWOA has been proposed for content-based image retrieval (NSMOWOA). This proposed method avoids the drawbacks in other non-dominated sorting

multi-objective methods that have been applied for content-based image retrieval via reducing the space and time complexity. The finding of the NSMOWOA revealed a good performance in content-based image retrieval problem according to recall and precision.

In Ref. [112], a new multi-objective WOA (WOAMOP) has been used that is purported to aim at multi-thresholding image segmentation. The hybrid between the Kapur's entropy and the Otsu's function was used by this proposed method as a fitness function. The WOAMOP applied to determine the best solution (threshold values) and next these thresholding values were used to separate the image. The results of experiments of the proposed method were contrasted and compared with the following four algorithms: Original WOA, FA, SSO, and FASSO. The WOAMOP outperformed to all the above-mentioned algorithms. It provides a more quick convergence with relatively lower processing time. In future, the WOAMOP can be used to other complex image segmentation problems like color images.

A novel or new hybrid system has been used or proposed that is based on a newly proposed named the MOWOA [113]. This system contains four modules as follows: a data preprocessing module, optimization module, forecasting module, and evaluation module. An effective and influential decomposing technique is also used to remove redundant or extra noise and extract the foremost and primary characteristics of wind speed data. In order to gain increased or high accuracy, and stability for wind speed prediction at the same time or simultaneously, and overcome the weaknesses of single objective optimization algorithms, the optimization module of the proposed MOWOA is applied to optimize the weights and thresholds of the Elman neural network utilized in the forecasting module. Lastly, the evaluation module, which contains hypothesis testing, evaluation criteria, and three experiments, is introduced to perform comprehensive evaluation on the system. The results show that the proposed MOWOA outperforms the two recently developed or novel MOALO and MODA algorithms, and that the proposed hybrid model is better in performance to all sixteen models used for comparison that shows its superior ability to generate or create forecasts in terms of forecasting accuracy and stability.

In Ref. [114], MOWOA has been utilized. The proposed algorithm was tested on some of the most challenging CEC 2009 benchmarks functions, and findings were compared with MOGWO, MOPSO, and MOEA/D. The proposed algorithm has increased optimization capabilities because of two mechanisms: Archive mechanism for non-dominated sorting and the second one was leader selection mechanism aimed at updating the position of whales according to archive. Performance analysis on six benchmark function shows superior results of MOWOA over the comparative algorithms in terms of diversity and generational distance. It can be applied to other real life engineering applications given better diversity and generational distance properties of MOWOA.

Variants of WOA with multi-objective are shown in Table 9.

According to the statistical result shown in Fig. 12, the binary and fuzzy methods have more percentage on WOA variants papers.

34. WOA applications in engineering optimization problems

WOA is utilized efficiently to backup applications of various fields such as engineering optimization problems. Formulating suitable problem and defining appropriate variables and objective functions are the major issues that can help efficient problem solving while dealing with engineering applications. Proper use of operators, parameters setting, and population illustration are the main concerns as regards the optimization algorithm. Here, a review of literature is carried out by categorizing the application areas. Various engineering optimization and applications of the WOA are categorized into the following classifications: Electrical engineering, Civil Engineering, Classification, Clustering, Image Processing, Mechanical Engineering, Control Engineering, Robot Path, Networks, Industrial Engineering, Task Scheduling, and other engineering applications. Table 10 demonstrates a summary of the applications of WOA in engineering optimization.

Table 9

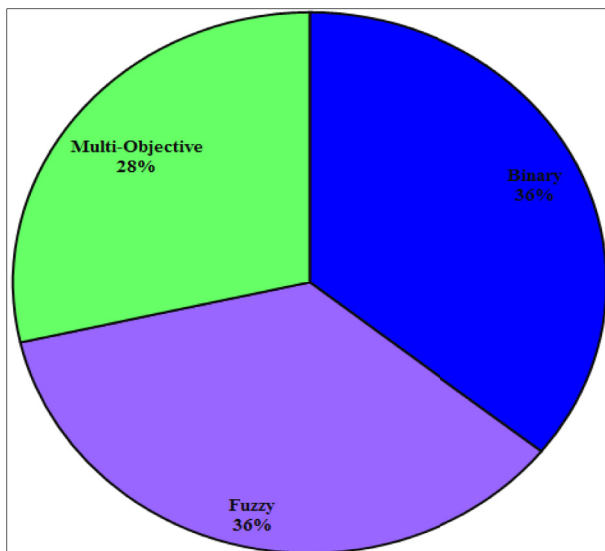
Variants of WOA with multi-objective.

Ref	Proposed Algorithm	Description	Application	Results
[111]	MOWOA	The main idea in the proposed method is the feature selection	content-based image retrieval	good performance in content-based image retrieval problem in terms of recall and precision
[112]	WOAMOP	The proposed method uses the whale algorithm to optimize threshold, and then uses this thresholding value to split the image.	Multilevel Thresholding Segmentation	better performance and provided faster convergence with a relatively lower processing time
[113]	MOWOA	An effective decomposing technique is also applied to eliminate redundant noise and extract the primary characteristics of wind speed data.	wind speed forecasting	superior ability to generate forecasts in terms of forecasting accuracy and stability
[114]	MOWOA	The proposed algorithm has improved optimization capabilities due to two mechanisms: Archive mechanism for non-dominated sorting and another was leader selection mechanism to update the position of whales according to archive	six benchmark unconstraint	accurate convergence

Table 10

A summary of the WOA applications in engineering optimization.

Category	Problems/Applications	Refs	Year
Electrical Engineering	Voltage Source Inverters	[118]	2018
	Economic Load Dispatch	[119]	
	Solar Photovoltaic	[120]	
	Photovoltaic Power Systems	[121]	2017
	Optimal Power Flow	[122]	
	Visible Light Communication	[123]	
	Economic Load Dispatch	[124]	
	Hydrothermal Scheduling	[125]	
	Photovoltaic Model Identification	[126]	
	Voltage Source Inverter	[127]	
	Radial Distribution Systems	[128]	
	Distribution Networks	[129]	
	Load Frequency Control	[130]	2016
	Radial Distribution Systems	[131]	
	Power System Stability	[132]	
	Thyristor Controlled Series Compensator	[133]	
	Power System	[134]	
	Distribution Networks	[135]	
	Economic Dispatch Problem	[136]	
	Distribution Systems	[137]	
	Economic Dispatch Problem	[138]	
	Economic Dispatch Problem	[139]	
Civil Engineering	Distribution Systems	[140]	2018
	fiber-reinforced polymer sheets	[141]	
	Sizing Optimization of Skeletal Structures	[142]	
	Sizing Optimization of Skeletal Structures	[143]	
Classification	Building Information Modeling	[144]	2016
	Feature Selection	[145]	
	Feature Selection	[146]	
	feature selection for E-mail Classification	[147]	
Clustering	Feature Selection	[148]	2018
	Feature Selection	[149]	
	Data clustering- UCI machine learning	[150]	
	Big data	[151]	
Image Processing	luster head section in Internet of Things	[152]	2017
	Data clustering- UCI machine learning	[153]	
	Liver segmentation in MRI images	[154]	
	Multilevel segmentation	[155]	
Mechanical Engineering	Image thresholding	[156]	2018
	power generation systems	[157]	
	Permanent Magnet Synchronous Motor	[158]	
	automatic generation control	[159]	
Control Engineering	fault location estimation in power systems	[160]	2017
	Orthogonal matching pursuit	[161]	
	Optimal Control Problems	[162]	
	Rendezvous Orbit Design	[163]	
Robot Path Networks	Aero-engine performance optimization	[164]	2017
	mobile robot path	[165]	
	Maximizing Lifetime of Wireless Sensor Networks	[166]	
	Optimal Vehicle path	[167]	
Industrial Engineering	Energy-Efficient Cluster Head Selection	[168]	2018
	Economic order quantity	[169]	
	Task Scheduling in Cloud Computing	[170]	
	Task Scheduling in Cloud Computing	[171]	
Others Problems	Task Scheduling in Cloud Computing	[172]	2017
	Task Scheduling in Cloud Computing	[173]	
	Flexible Flow Shop Scheduling with Setup Times	[174]	
	unequally spaced array	[175]	
	Cryptanalysis in Merkle-Hellman Cryptosystem	[176]	2018
	prioritize the software requirements	[177]	
	Clinical data analysis	[178]	
	Offset voltage minimization based circuit sizing of CMOS operational amplifier	[179]	
	Function Optimization	[180]	2016
	mapping of chemical gas concentration and localizing the source	[181]	
	Link Prediction	[182]	
	signal processing	[183]	
	Global Optimization	[184]	2016
	Parallel analysis	[185]	
	Global Optimization	[185]	

**Fig. 12.** Percentage of WOA variants papers.

Meta-heuristics algorithms is utilized the cooperation as an approach for the swapping of information between the algorithms involved in finding the solution of the problem. This swapping of information has, as

the main goal, to guide the search for the most promising areas of the solution space. Meta-heuristic algorithms play important roles in science, management and industry, given that many issues in the real world can essentially be shaped as optimization tasks. “Traditional” mathematical programming methods (e.g., gradient-based methods) are no longer completely effective in solving complex optimization problems characterized by multi-modality, discontinuity, and noise.

With the advances in science and technology, optimization problems in engineering come to be increasingly intricate, and consequently the calculation of gradients becomes more heavy and difficult because of the implicit dependence on design variables and the needful of a large amount of computational memory. In addition, new engineering problems commonly have some features in common such as lacking an explicit mathematical formulation, non-continuous definition domains, and multimodal characteristics. Hence, classical gradient-based optimization methods are unfavorable and inadequate to solve today's complicated engineering problems, which urges the creation and development of new optimization strategies. Compared to traditional methods, meta-heuristic algorithms usually do not require gradient problem information and have superior features such as simple concepts, easy implementation, insensitivity to select initial values, stable and fast convergence and high computational efficiency. Researches demonstrate that meta-heuristics are more efficient at tackling NP-hard problems and become powerful tools of solving complicated global optimization problems.

Fig. 13 indicates numbers of papers of WOA application in different filed of engineering optimization problems. On the whole, many researchers have previously carried out a lot of enhancements and adjustments. In short, each and various aims or objectives of study require different methods to obtain the desired results. The solution of every problem cannot be limited to just one method. Hence, the development of the solution will be continuously expanded.

5. Discussion

Global numerical optimization is referred to a procedure of gaining optimal solutions of a system or mathematical model by maximizing or minimizing an objective function. Due to the enhancing complexity of optimization tasks used in various real-world problems, the development of efficient optimization techniques is becoming more critical and important and urgent than before. Across the previous decades, various or different heuristic methodologies have been used according to different phenomena in biology or nature. Given the number of candidate solutions, these algorithms can be divided into two sub-categories of individual-based algorithms and population-based algorithms. Substantial gradient information is needed by the former that is individual-based algorithms and it sometimes; it begins with a single random point,

enhanced over the course of generations. Less computation is required by these algorithms and a good strategy is provided to find the global optimal agent for simple models. However, they usually have demerits such as derivation-based mechanism, premature convergence, and unstable search in complex environments. Regarding the latter, the optimization process has a tendency to begin with a set of random solutions, sampled from the solution space and increased from generation to generation. It would be really helpful and useful to apply this method to avoid local optima. In addition, the transfer of information between the solutions also assists the algorithm come over different difficulties of different search spaces.

The classification of the solving process into two main milestones is the main benefit of population-based stochastic search methods [194]: diversification and intensification. The former or diversification refers to a process in which the population agents have a tendency to be altered more frequently and investigate the promising regions as vast as possible. In contrast, the search agents are guided or leaded by the latter to converge toward the best solution gained in the diversification phase. The result of focusing on intensification is fast convergence, however, higher local optima avoidance is the result of emphasizing diversification. So, a good technique should try to achieve a good balance between them. More recently, meta-heuristic methods have performed remarkably in solving a wide variety of real-world optimization problems due to the plus points such as simplicity, flexibility, derivation-free mechanism, and local optima avoidance. That is why the population-based algorithms have expended to different areas speedily. Scholars or researchers have used many successful and efficient optimization methods for the sake of being inspired by nature.

The success of metaheuristics is explained by numerous reasons. First, the use of stochastic operators allows meta-heuristics to escape from local optima and converge to approximate global optima. The objective here is not to find the exact optimal solution of the problem at hand but to find solution(s) of good quality within a reasonable computational time. The key element in order to achieve such an objective depends on the suitable trade-off between exploration and exploitation. Exploration aims to identify the most promising regions in a search space through the diversification of search directions. Therefore, the search is locally intensified by the exploration in a promising region to find solutions of high quality. The better the performance of a specific meta-heuristic algorithm in balancing these two operations, the better its performance will be. The existing meta-heuristics obtain the balance between these two operations differently. They might be more adapted for exploration or for exploitation. Two or more algorithms are combined by hybrid metaheuristic to take advantages from each while avoiding as much as possible their drawbacks. The second reason for the success of meta-heuristics is because of their generality and ease of use and tune in practice.

Hybridization of meta-heuristics takes advantage of the power point in each algorithm. It can be done between different techniques in low or high levels depending on the degree of interference between both parts of the algorithms, i.e. high-level hybridization demonstrates that low interference between the internal actions of hybridized algorithms while low-level hybridization means that only a meta-heuristic function is diffused to another meta-heuristic. During the search process, the hybridized algorithms may collaboratively exchange information or integrated with a master one that operates the search method. Besides, the algorithm execution sequence must be taken into account.

The nature-inspired optimization methods are commonly known as meta-heuristic algorithms, in an abstract sense. Meta-heuristics are high-level methods that guide the search agents to progressively improve the totally solution. The solution is therefore optimized by checking across some random elements and possibilities, in the hope of finding a better quality solution, while the candidate solution is inherited from one stochastic iteration to the next. In contrast to deterministic methods, meta-heuristics may not be an absolute best solution. But quite often they produce adequately good solution in most cases within an equitable

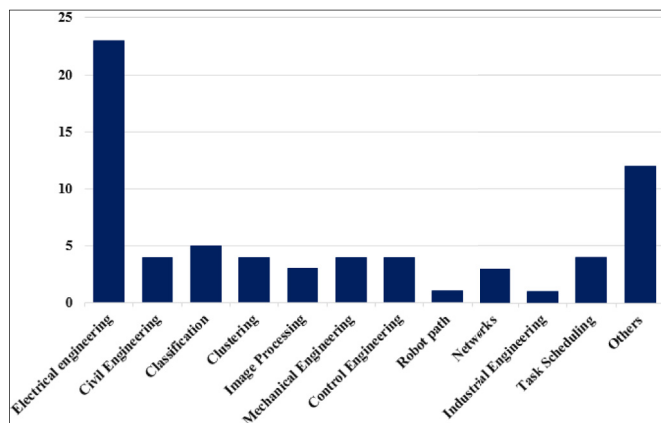


Fig. 13. Graph showing of papers of WOA application in various filed of engineering optimization problems.

amount of time.

The meta-heuristic have been utilized extensively to solve various optimization problems. However, for sophisticated cases or real life cases, most of the algorithms still suffer from local optima trapped and fail to reach the global optimum solution. This is by reason of the lack exploration (diversification) component inside the algorithm. Several diversification methods were applied to propagate the performance of distinguished meta-heuristic algorithms and help counteract the drawbacks. These methods are hybridization and elitism [198,199].

WOA has become an important tool for optimization and other complicated problem solving, like other meta-heuristic algorithms. It is an intelligent computational method to find global solution multidimensional functions based on minimum and maximum. The WOA has ordinary principle which can be simply used in a large range of fields. The results reported in this review provide evidence of performance attainment of the WOA in terms of both performance and quality of solution. This assertion is established based on reviewing the results of comparisons between the WOA and some other well-known evolutionary algorithms.

Similar to different meta-heuristic algorithms, WOA has both advantages and some unavoidable shortcomings. Although there is no convergence proof for this algorithm, the results reported in this study confirms its competitiveness over other evolutionary algorithms in terms of convergence rate. Table 11 summarizes the strength and weaknesses of WOA. Advantages and disadvantages of WOA shown in Table 11.

What has been observed in recent years further complicates the discussion on meta-heuristic based algorithms; generally, these algorithms seem to provide varying performance depending upon the sensitivity (skills, expertise, ingenuity, etc.) of the designer in algorithmic fine tuning. In order to maximize algorithmic performance, an instance specific fine tuning might be required. The variability in the results of a metaheuristic presents at least two major drawbacks: (I) On one hand, the problem of reproducibility of results arises; given such factors as implementation skills, special usage of data structures, and ability in parameter settings, among others, It is common knowledge that a proposed algorithm implemented by two different researchers can yield totally different results. That is why, comparing and thoroughly appraising a meta-heuristic and its performance is hard. (II) On the other hand, it is predicted that a problem of performance maximization occurs: one of the commonalities of virtually all proposed metaheuristic paradigms is that they are characterized by a considerable number of parameters whose value(s) strongly affect the overall performance of the

Table 11
Advantages and disadvantages of WOA.

Advantages	Disadvantages
✓Combining with different algorithms is incredibly comfortable	X The original algorithm has been designed for continuous search space
✓A good rate of convergence	X Too many parameter tuning
✓Rapid discovery of good solutions	X May result in premature convergence
✓Appropriate for various types of optimization problems	X No theoretical convergent property
✓Efficient global approach to search	X Probability distribution changes by iterations
✓Appropriate for both continuous and discrete search space	
✓Strong neighborhood search property	
✓Flexibility, robustness and scalability are found notable features	
✓Powerful in handling large number of decision variables	
✓Have higher probability and efficiency in finding the global optima	
✓Lower probably of trapping into local optima	
✓Less dependency on initial solutions	
✓WOA is easy in its concept and coding implementation compared to other heuristic optimization techniques	
✓Reasonable computational time	

algorithm itself. It is prevalent knowledge that the fine tuning of algorithmic parameters is not just problem-specific, but it is instance-specific. It means even for a given problem, parameter values should be fine-tuned and adjusted based on instance specific information (e.g., instance size, distribution of its values, etc.).

One of the difficulties associated with WOA is that a study of the probabilistic convergence attributes of WOA is required to better understand the algorithm. The problem of premature convergence in the WOA often leads the search process to be trapped in a local optimum. This problem often occurs when the population diversity decreases and the whales cannot escape from a local optimal. In addition, there are great potentials for engineers to use the advantages WOA encountering involved industry problems.

Fig. 14 shows the number of WOA articles published in various methods like hybrid, and altered methods in addition to the applications according to the publication year.

From the literature, there are trending patterns occurred from the year 2016–2018. Fig. 14 shows various type of hybridization, improved, variants, and applications in terms of the total number of studies from each year. The years shown clearly demonstrate that the WOA was applied into different methods. As stated in the figure, 2017 shows the highest use of the WOA compared to other years. This clearly shows that the use of the WOA in the optimization field has yield interest after 2 years.

6. Conclusion

A literature review of the WOA is presented in this paper. This study is objectively aimed at summarizing the overview and the application of WOA in all categories reviewed. It is confirmed that according to the studies in the literature standard WOA in combination with chaotic map was the main technique used in WOA. In addition, we also obtained information from the reviews that the applications prove the wide application of WOA in engineering, classification, clustering, image processing, task scheduling, Networking, and other problems. Furthermore, previous researchers have found that WOA could solve the optimization problems such as engineering design optimization, unconstrained optimization problems, and pattern recognition problems. In short, WOA is very efficient in all the test problems according to the review of literature. In conclusion, as there is still room or space for performance, the WOA could be expanded into various hybridization, improved, and variants based on the necessity of the problems. Therefore, the results of this study could be applied by other future and interested researchers to a different process or for improvement purpose by taking into attention on the applications, merits or demerits of other

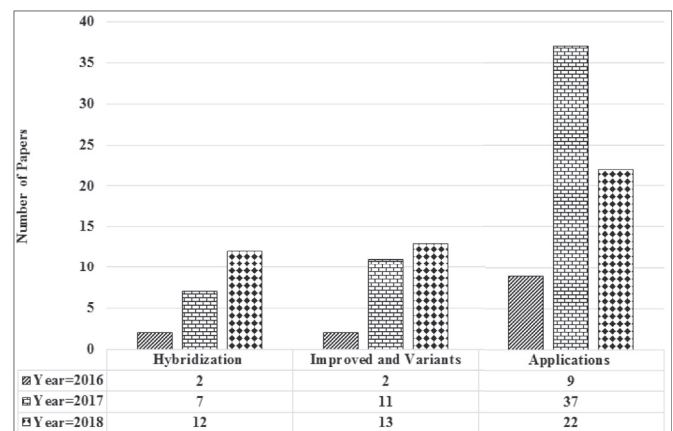


Fig. 14. The number of WOA articles published in different methods such as hybrid, and improved and variants methods plus the applications based on the publication year.

system proposed by previous researchers. For future works, we recommend to exploit and improve the WOA with other meta-heuristic algorithms for further benefits for solving continuous optimization problems. In the future, we will focus on the following aspects:

- 1) Utilizing WOA to solve MOPs.
- 2) Modifying WOA to deal with real-world optimization problems, especially the discrete optimization problems and the NP-hard problems.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.swevo.2019.03.004>.

References

- [1] H. Shayanfar, F.S. Gharehchopogh, Farmland fertility: a new metaheuristic algorithm for solving continuous optimization problems, *Appl. Soft Comput.* 71 (2018) 728–746.
- [2] M.A. Elhosseini, A.Y. Haikal, M. Badawy, N. Khashan, Biped robot stability based on an A-C parametric Whale Optimization Algorithm, *Journal of Computational Science* 31 (2019) 17–32.
- [3] Yonghao Miao, Ming Zhao, Viliam Makis, Jing Lin, Optimal swarm decomposition with whale optimization algorithm for weak feature extraction from multicomponent modulation signal, *Mech. Syst. Signal Process.* 122 (2019) 673–691.
- [4] J. Ghahremani-Nahr, R. Kian, E. Sabet, A robust fuzzy mathematical programming model for the closed-loop supply chain network design and a whale optimization solution algorithm, *Expert Syst. Appl.* 116 (2019) 454–471.
- [5] B. He, Y. Huang, D. Wang, B. Yan, D. Dong, A parameter-adaptive stochastic resonance based on whale optimization algorithm for weak signal detection for rotating machinery, *Measurement* 136 (March 2019) 658–667.
- [6] Dalia Yousri, Dalia Allam, M.B. Eteiba, Chaotic whale optimizer variants for parameters estimation of the chaotic behavior in Permanent Magnet Synchronous Motor, *Appl. Soft Comput.* 74 (2019) 479–503.
- [7] P.R. Singh, M.A. Elaziz, S. Xiong, Modified Spider Monkey Optimization based on Nelder–Mead method for global optimization, *Expert Syst. Appl.* 110 (2018) 264–289.
- [8] L. Ingber, Very fast simulated re-annealing, *Math. Comput. Model.* 12 (8) (1989) 967–973.
- [9] A.H. Gandomi, X.S. Yang, Chaotic bat algorithm, *Journal of Computational Science* 5 (2) (2014) 224–232.
- [10] N.A. Kallioras, N.D. Lagaros, D.N. Avtzis, Pity beetle algorithm – a new metaheuristic inspired by the behavior of bark beetles, *Adv. Eng. Software* 121 (2018) 147–166.
- [11] S. Arora, S. Singh, An improved butterfly optimization algorithm with chaos, *J. Intell. Fuzzy Syst.* 32 (1) (2017) 1079–1088.
- [12] O.E. Turgut, Thermal and economical optimization of a shell and tube evaporator using hybrid backtracking search-sine-cosine algorithm, *Arabian J. Sci. Eng.* 42 (5) (2017) 2105–2123.
- [13] H. Ma, S. Shen, M. Yu, Z. Yang, H. Zhou, Multi-population techniques in nature inspired optimization algorithms: a comprehensive survey, *Swarm Evol. Comput.* 44 (January 2019) 365–387.
- [14] R.M. Rizk-Allah, Hybridizing sine cosine algorithm with multi-orthogonal search strategy for engineering design problems, *Journal of Computational Design and Engineering* 5 (2) (2018) 249–273.
- [15] D. Tian, Z. Shi, MPSO: modified particle swarm optimization and its applications, *Swarm and Evolutionary Computation* 41 (2018) 49–68.
- [16] I. Boussaid, J. Lepagnot, P. Siarry, A survey on optimization metaheuristics, *Inf. Sci.* 237 (2013) 82–117.
- [17] W.G. Jackson, E. Ozcan, R.I. John, Move acceptance in local search metaheuristics for cross-domain search, *Expert Syst. Appl.* 109 (2018) 131–151.
- [18] G. Dhiman, V. Kumar, Emperor penguin optimizer: a bio-inspired algorithm for engineering problems, *Knowl. Base Syst.* 159 (November 2018) 20–50.
- [19] J. Holland, *Adaptation in Natural and Artificial Systems*, University of Michigan, Michigan, USA, 1975.
- [20] A. Bhattacharya, P.K. Chattopadhyay, Solving complex economic load dispatch problems using biogeography-based optimization, *Expert Syst. Appl.* 37 (5) (2010) 3605–3615.
- [21] A. Kaveh, M. Khayatizad, A new meta-heuristic method: Ray Optimization, *Comput. Struct.* 112–113 (2012) 283–294.
- [22] H. Abedinpourshotorban, S.M. Shamsuddin, Z. Beheshti, D.N.A. Jawawi, Electromagnetic field optimization: a physics-inspired metaheuristic optimization algorithm, *Swarm and Evolutionary Computation* 26 (2016) 8–22.
- [23] M. Yin, Y. Hu, F. Yang, X. Li, W. Gu, A novel hybrid K-harmonic means and gravitational search algorithm approach for clustering, *Expert Syst. Appl.* 38 (8) (2011) 9319–9324.
- [24] E. Pacini, C. Mateos, C.G. Garino, Distributed job scheduling based on Swarm Intelligence: a survey, *Comput. Electr. Eng.* 40 (1) (2014) 252–269.
- [25] D. Tian, Z. Shi, MPSO: modified particle swarm optimization and its applications, *Swarm and Evolutionary Computation* 41 (2018) 49–68.
- [26] J. Kennedy, R.C. Eberhart, Particle swarm optimization, in: *Proceedings of the IEEE International Conference on Neural Networks*, 1995, pp. 1942–1948.
- [27] D. Karaboga, An Idea Based on Honeybee Swarm for Numerical Optimization, Technical Report TR06, Erciyes University, Engineering Faculty, Computer Engineering Department, 2005.
- [28] X.S. Yang, *Nature-Inspired Meta-Heuristic Algorithms*, Luniver Press, 2008.
- [29] F. Fausto, E. Cuevas, A. Valdivia, A. Gonzalez, A global optimization algorithm inspired in the behavior of selfish herds, *Biosystems* 160 (2017) 39–55.
- [30] O. Hasançebi, T. Teke, O. Pekcan, A bat-inspired algorithm for structural optimization, *Comput. Struct.* 128 (2013) 77–90.
- [31] A.H. Gandomi, A.H. Alavi, S. Talatahari, Structural Optimization Using Krill Herd Algorithm, *Swarm Intelligence and Bio-Inspired Computation*, 2013, pp. 335–349.
- [32] S.C. Chu, P.W. Tsai, J.S. Pan, in: *Cat Swarm Optimization*, Pacific Rim International Conference on Artificial Intelligence PRICAI 2006: PRICAI 2006: Trends in Artificial Intelligence, 2006, pp. 854–858.
- [33] S. Mirjalili, S.M. Mirjalili, Andrew Lewis, grey wolf optimizer, *Adv. Eng. Software* 69 (2014) 46–61.
- [34] S. Mirjalili, The ant lion optimizer, *Adv. Eng. Software* 83 (2015) 80–98.
- [35] X. Qi, Y. Zhu, H. Zhang, A new meta-heuristic butterfly-inspired algorithm, *Journal of Computational Science* 23 (2017) 226–239.
- [36] M.D. Li, H. Zhao, X.W. Weng, T. Han, A novel nature-inspired algorithm for optimization: Virus colony search, *Adv. Eng. Software* 92 (2016) 65–88.
- [37] M. Jain, V. Singh, A. Rani, A novel nature-inspired algorithm for optimization: Squirrel search algorithm, *Swarm Evol. Comput.* 44 (February 2019) 148–175.
- [38] Y.J. Zheng, Water wave optimization: a new nature-inspired metaheuristic, *Comput. Oper. Res.* 55 (2015) 1–11.
- [39] V.M. Nakarajan, M.M. Noel, Galactic Swarm Optimization: a new global optimization metaheuristic inspired by galactic motion, *Appl. Soft Comput.* 38 (2016) 771–787.
- [40] B. Dogan, T. Olmez, A new metaheuristic for numerical function optimization: Vortex Search algorithm, *Inf. Sci.* 293 (2015) 125–145.
- [41] A. Askarzadeh, A novel metaheuristic method for solving constrained engineering optimization problems: Crow search algorithm, *Comput. Struct.* 169 (2016) 1–12.
- [42] S. Mirjalili, A. Lewis, The whale optimization algorithm, *Adv. Eng. Software* 95 (2016) 51–67.
- [43] P.R. Hof, E.V.D. Gucht, Structure of the cerebral cortex of the humpback whale, *Megaptera novaeangliae* (Cetacea, Mysticeti, Balaenopteridae), *Anat. Rec.* 290 (2007) 1–31.
- [44] W.A. Watkins, W.E. Schevill, Aerial observation of feeding behavior in four baleen whales: *Eubalaena glacialis*, *Balaenoptera borealis*, *Megaptera novaeangliae*, and *Balaenoptera physalus*, *J. Mammal.* (1979) 155–163.
- [45] J.A. Goldbogen, A.S. Friedlaender, J. Calambokidis, M.F. Mckenna, M. Simon, D.P. Nowacek, Integrative approaches to the study of baleen whale diving behavior, feeding performance, and foraging ecology, *Bioscience* 63 (2013) 90–100.
- [46] R. Senkerik, I. Zelinka, M. Pluhacek, A. Viktorin, J. Janostik, Z.K. Oplatkova, Randomization and Complex Networks for Meta-Heuristic Algorithms, *Evolutionary Algorithms, Swarm Dynamics and Complex Networks*, 2018, pp. 177–194.
- [47] I. Fister, D. Strnad, X.S. Yang, I. Fister, Adaptation and Hybridization in Nature-Inspired Algorithms, *Adaptation and Hybridization in Computational Intelligence*, 2015, pp. 3–50.
- [48] N.M. Laskar, K. Guha, I. Chatterjee, S. Chanda, K.L. Baishnab, P.K. Paul, HWPSON: a new hybrid whale-particle swarm optimization algorithm and its application in electronic design optimization problems, *Appl. Intell.* (2018) 1–27.
- [49] I.N. Trivedi, P. Jangir, A. Kumar, N. Jangir, R. Totlani, A novel hybrid PSO-WOA algorithm for global numerical functions optimization, *Advances in Computer and Computational Sciences* 554 (2018) 53–60.
- [50] M.A. Basset, G. Manogaran, D.E. Shahat, S. Mirjalili, A hybrid whale optimization algorithm based on local search strategy for the permutation flow shop scheduling problem, *Future Gener. Comput. Syst.* 85 (2018) 129–145.
- [51] N. Singh, H. Hachimi, A new hybrid whale optimizer algorithm with mean strategy of grey wolf optimizer for global optimization, *Math. Comput. Appl.* (2018) 1–32.
- [52] A. Kaveha, M.R. Moghaddam, A Hybrid WOA-CBO Algorithm for Construction Site Layout Planning Problem, vol. 25, Sharif University of Technology, 2018, pp. 1094–1104 (3).
- [53] A. Kaveh, V.R. Mahdavi, Colliding bodies optimization: a novel meta-heuristic method, *Comput. Struct.* 139 (2014) 18–27.
- [54] Z. Xu, Y. Yu, H. Yachi, J. Ji, Y. Todo, S. Gao, A novel memetic whale optimization algorithm for optimization, in: *International Conference on Swarm Intelligence*, ICSI 2018: Advances in Swarm Intelligence, vol. 10941, 2018, pp. 384–396.
- [55] M.M. Mafarja, S. Mirjalili, Hybrid Whale Optimization Algorithm with simulated annealing for feature selection, *Neurocomputing* 260 (2017) 302–312.
- [56] M.A. El Aziz, A.A. Ewees, A.E. Hassanien, Whale optimization algorithm and moth-flame optimization for multilevel thresholding image segmentation, *Expert Syst. Appl.* 83 (2017) 242–256.
- [57] S. Mirjalili, Moth-flame optimization algorithm: a novel nature-inspired heuristic paradigm, *Knowl. Base Syst.* 89 (2015) 228–249.
- [58] A.N. Jadhav, N. Gomathi, WGC: hybridization of exponential grey wolf optimizer with whale optimization for data clustering, *Alexandria Eng. J.* 57 (3) (September 2018) 1569–1584.
- [59] F.A. Mohamed, M.A. Nasser, K. Mahmoud, S. Kamel, Accurate economic dispatch solution using hybrid whale-wolf optimization method, in: *Nineteenth*

- International Middle East Power Systems Conference (MEPCON), IEEE, 2017, pp. 922–927.
- [60] B. Bentouati, L. Chaib, S. Chettih, A hybrid whale algorithm and pattern search technique for optimal power flow problem, in: 8th International Conference on Modelling, Identification and Control (ICMIC), 2016, pp. 1048–1053.
- [61] S. Khalilpourazary, S. Khalilpourazary, SCWOA: an efficient hybrid algorithm for parameter optimization of multi-pass milling process, *Journal of Industrial and Production Engineering* (2018) 1–14. Taylor & Francis.
- [62] S. Mirjalili, SCA: a Sine Cosine Algorithm for solving optimization problems, *Knowl. Base Syst.* 96 (2016) 120–133.
- [63] S.T. Revathi, N. Ramaraj, S. Chithra, Brain storm-based Whale Optimization Algorithm for privacyprotected data publishing in cloud computing, *Clust. Comput.* (2018) 110.
- [64] Y. Shi, Brain storm optimization algorithm, in: *International Conference in Swarm Intelligence, ICSI 2011: Advances in Swarm Intelligence*, vol. 6728, Springer, 2011, pp. 303–309.
- [65] G. Villarrubia, J.F.D. Paz, P. Chamoso, Fernando De la Prieta, Artificial neural networks used in optimization problems, *Neurocomputing* 272 (2018) 10–16.
- [66] Virupakshappa, B. Amarapur, Computer-aided diagnosis applied to MRI images of brain tumor using cognition based modified level set and optimized ANN classifier, *Multimed. Tool. Appl.* (2018) 1–29.
- [67] D. Parambanchary, V.M. Rao, WOA-NN: a decision algorithm for vertical handover in heterogeneous networks, *Wireless Network* (2018) 1–16.
- [68] J. Rajeshkumar, K. Kousalya, Diabetes data classification using whale optimization algorithm and backpropagation neural network, *Int. Res. J. Pharm.* 8 (11) (2017) 219–222.
- [69] I. Aljarah, H. Faris, S. Mirjalili, Optimizing connection weights in neural networks using the whale optimization algorithm, *Soft Computing* 22 (Issue 1) (2016) 1–15.
- [70] B. Ghaddar, J.N. Sawaya, High dimensional data classification and feature selection using support vector machines, *Eur. J. Oper. Res.* 265 (3) (2018) 993–1004.
- [71] N.P. Karlekar, N. Gomathi, OW-SVM: ontology and whale optimization-based support vector machine for privacy-preserved medical data classification in cloud, *Int. J. Commun. Syst.* 31 (12) (2018).
- [72] A.M. Al-Zoubi, H. Faris, J. Alqatawna, M.A. Hassonah, Evolving Support Vector Machines using Whale Optimization Algorithm for spam profiles detection on online social networks in different lingual contexts, *Knowl. Base Syst.* 153 (2018) 91–104.
- [73] A. Tharwat, Y.S. Moemen, A.E. Hassanien, Classification of toxicity effects of biotransformed hepatic drugs using whale optimized support vector machines, *J. Biomed. Inform.* 68 (2017) 132–149.
- [74] R.K. Saidala, N.R. Devarakonda, Bubble-net hunting strategy of whales based optimized feature selection for e-mail classification, in: *2nd International Conference for Convergence in Technology (I2CT)*, 2017, pp. 626–631.
- [75] E. Hancer, B. Xue, M. Zhang, Differential evolution for filter feature selection based on information theory and feature ranking, *Knowl. Base Syst.* 140 (2018) 103–119.
- [76] H. Ghimatgar, K. Kazemi, M.S. Helfroush, A. Aarabi, An improved feature selection algorithm based on graph clustering and ant colony optimization, *Knowl. Base Syst.* 159 (November 2018) 270–285.
- [77] L.K. Panwar, S. ReddyK, A. Verma, B.K. Panigrahi, R. Kumar, Binary Grey Wolf Optimizer for large scale unit commitment problem, *Swarm and Evolutionary Computation* 38 (2018) 251–266.
- [78] Y.K. Wu, H.Y. Chang, S.M. Chang, Analysis and comparison for the unit commitment problem in a large-scale power system by using three meta-heuristic algorithms, *Energy Procedia* 141 (2017) 423–427.
- [79] Y. He, X. Wang, Group theory-based optimization algorithm for solving knapsack problems, *Knowl. Base Syst.* (August 2018).
- [80] E. Ulker, V. Tongur, Migrating birds optimization (MBO) algorithm to solve knapsack problem, *Procedia Computer Science* 111 (2017) 71–76.
- [81] A.G. Hussien, A.E. Hassanien, E.H. Houssein, S. Bhattacharyya, M. Amin, S-shaped binary whale optimization algorithm for feature selection, *Recent Trends in Signal and Image Processing* 727 (2019) 79–87.
- [82] S. Reddy K, L. Panwar, B.K. Panigrahi, R. Kumar, Binary Whale Optimization Algorithm: a New Metaheuristic Approach for Profit-Based Unit Commitment Problems in Competitive Electricity Markets, *Engineering Optimization*, Taylor & Francis, 2018, pp. 1–22.
- [83] H.F. Eid, Binary whale optimisation: an effective swarm algorithm for feature selection, *Int. J. Metaheuristics (IJMHeur)* 7 (1) (2018).
- [84] A.G. Hussien, E.H. Houssein, A.E. Hassanien, A binary whale optimization algorithm with hyperbolic tangent fitness function for feature selection, in: *2017 Eighth International Conference on Intelligent Computing and Information Systems (ICICIS)*, IEEE, 2017, pp. 166–172.
- [85] H. Zamani, M.H. Nadimi-Shahraki, Feature selection based on whale optimization algorithm for diseases diagnosis, *Int. J. Comput. Sci. Inf. Secur.* 14 (No. 9) (2016) 1243–1247.
- [86] G. Kaur, S. Arora, Chaotic whale optimization algorithm, *Journal of Computational Design and Engineering* 5 (3) (2018) 275–284.
- [87] P. Yuan, C. Guo, Q. Zheng, J. Ding, Sidelobe suppression with constraint for MIMO radar via chaotic whale optimisation, *Electron. Lett.* 54 (5) (2018) 311–313.
- [88] G.I. Sayed, A new chaotic whale optimization algorithm for features selection, *J. Classif.* (2018) 1–45.
- [89] D. Oliva, M.A.E. Aziz, A.E. Hassanien, Parameter estimation of photovoltaic cells using an improved chaotic whale optimization algorithm, *Appl. Energy* 200 (2017) 141–154.
- [90] D. Prasad, A. Mukherjee, V. Mukherjee, Transient Stability Constrained Optimal Power Flow Using Chaotic Whale Optimization Algorithm, *Handbook of Neural Computation*, 2017, pp. 311–332.
- [91] D. Prasad, A. Mukherjee, G. Shankar, V. Mukherjee, Application of chaotic whale optimisation algorithm for transient stability constrained optimal power flow, *IET Sci. Meas. Technol.* 11 (8) (2017) 1002–1013.
- [92] W.Z. Sun, J.S. Wang, Elman neural network soft-sensor model of conversion velocity in polymerization process optimized by chaos whale optimization algorithm, *IEEE Access* 5 (2017) 13062–13076.
- [93] R. Sivalingam, S. Chinnamuthu, S.S. Dash, A modified whale optimization algorithm based adaptive fuzzy logic PID controller for load frequency control of autonomous power generation systems, *Journal for Control, Measurement, Electronics, Computing and Communications* (2018) 410–421. Taylor & Francis.
- [94] W.R.S. Emmanuel, S.J. Minija, Fuzzy Clustering and Whale-Based Neural Network to Food Recognition and Calorie Estimation for Daily Dietary Assessment, *Sadhana*, 2018, pp. 1–19.
- [95] H.V. Patil, S.D. Shirbahadurkar, FWFusion: Fuzzy Whale Fusion Model for MRI Multimodal Image Fusion, 2018, pp. 1–16.
- [96] S.K. Harikarthik, V. Palanisamy, P. Ramanathan, Optimal test suite selection in regression testing with testcase prioritization using modified Ann and Whale optimization algorithm, *Clust. Comput.* (2017) 1–10.
- [97] A.E. Hassanien, M.A. Elfattah, S. Aboulenin, G. Schaefer, S.Y. Zhu, I. Korovin, Historic handwritten manuscript binarisation using whale optimisation, in: *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, IEEE, 2016, pp. 3842–3846.
- [98] X.S. Yang, D. Suash, Cuckoo search via levy flights, in: *World Congress Nature Biologically Inspired Computer (NaBIC)*, IEEE Publication, New York, NY, USA, 2009, pp. 210–214.
- [99] A.F. Kamaruzaman, A.M. Zain, S.M. Yusuf, A. Udin, Levy flight algorithm for optimization problems-A literature review, *Appl. Mech. Mater.* 421 (2013) 496–501.
- [100] Y. Ling, Y. Zhou, Q. Luo, Levy flight trajectory-based whale optimization algorithm for global optimization, *IEEE Access* 5 (2017) 6168–6186.
- [101] Y. Sun, X. Wang, Y. Chen, Z. Liu, A modified whale optimization algorithm for large-scale global optimization problems, *Expert Syst. Appl.* 114 (December 2018) 563–577.
- [102] M. Abdel-Basset, L. Abde-Fatah, A.K. Sangaiah, An improved Levy based whale optimization algorithm for bandwidth-efficient virtual machine placement in cloud computing environment, *Clust. Comput.* (2018) 1–16.
- [103] M. Abdel-Basset, D. El-Shahat, A.K. Sangaiah, A modified nature inspired meta-heuristic whale optimization algorithm for solving 0–1 knapsack problem, *International Journal of Machine Learning and Cybernetics* (2017) 1–20.
- [104] Q. Lin, Y. Ma, J. Chen, Q. Zhu, F. Chen, An adaptive immune-inspired multi-objective algorithm with multiple differential evolution strategies, *Inf. Sci.* 430–431 (2018) 46–64.
- [105] G. Dhiman, V. Kumar, Multi-objective spotted hyena optimizer: a Multi-objective optimization algorithm for engineering problems, *Knowl. Base Syst.* 150 (2015) 175–197.
- [106] K. Deb, *Advances in Evolutionary Multi-Objective Optimization*, Springer Berlin Heidelberg, Berlin, Heidelberg, 2012, pp. 1–26.
- [107] K. Deb, A. Pratap, S. Agarwal, T. Meyarivan, A fast and elitist multi-objective genetic algorithm: nsga-ii, *IEEE Trans. Evol. Comput.* 6 (2) (2002) 182–197.
- [108] C.A. Coello Coello, M. S. Lechuga Mopso, A proposal for multiple objective particle swarm optimization, in: *Proceedings of the Evolutionary Computation on 2002. CEC '02. Proceedings of the 2002 Congress*, IEEE Computer Society, CEC'02, vol. 02, 2002, pp. 1051–1056. Washington, DC, USA.
- [109] Q. Zhang, H. Li, MOEA/D: a multi-objective evolutionary algorithm based on decomposition, *IEEE Trans. Evol. Comput.* 11 (6) (2007) 712–731.
- [110] D. Angus, C. Woodward, Multiple objective ant colony optimisation, *Swarm Intelligence* 3 (1) (2009) 69–85.
- [111] M.A.E. Aziz, A.A. Ewees, A.E. Hassanien, Multi-objective whale optimization algorithm for content-based image retrieval, *Multimed. Tool. Appl.* (2018) 1–38.
- [112] M.A.E. Aziz, A.A. Ewees, A.E. Hassanien, M. Mudhs, S. Xiong, *Advances in Soft Computing and Machine Learning in Image Processing*, vol. 730, 2018, pp. 23–39.
- [113] J. Wang, P. Du, T. Niu, W. Yang, A novel hybrid system based on a new proposed algorithm-Multi-Objective Whale Optimization Algorithm for wind speed forecasting, *Appl. Energy* 208 (2017) 344–360.
- [114] I.R. Kumawat, S.J. Nanda, R.K. Maddila, Multi-objective whale optimization, in: *TENCON 2017 - 2017 IEEE Region 10 Conference*, IEEE, 2017, pp. 2747–2752.
- [115] H.R. Tizhoosh, Opposition-based learning: a new scheme for machine intelligence, in: *International Conference on Computational Intelligence for Modelling, Control and Automation and International Conference on Intelligent Agents, Web Technologies and Internet Commerce (CIMCA-IAWTIC'06)*, vol. 1, 2005, pp. 695–701.
- [116] M.A. Elaziz, D. Oliva, Parameter estimation of solar cells diode models by an improved opposition-based whale optimization algorithm, *Energy Convers. Manag.* 171 (2018) 1843–1859.
- [117] S.H. Alamri, Y.A. Alsariera, K.Z. Zamli, Opposition-based whale optimization algorithm, *Advanced Science Letters*, *Advanced Science Letters* 24 (10) (2018) 7461–7464.
- [118] G. Nalcaiz, M. Ermis, Selective harmonic elimination for three-phase voltage source inverters using whale optimizer algorithm, in: *5th International Conference on Electrical and Electronic Engineering (ICEEE)*, 2018, pp. 1–6.
- [119] F.A. Mohamed, M. Abdel-Nasser, K. Mahmoud, S. Kamel, Economic dispatch using stochastic whale optimization algorithm, in: *International Conference on Innovative Trends in Computer Engineering (ITCE)*, 2018, pp. 19–24.

- [120] G. Xiong, J. Zhang, D. Shi, Y. He, Parameter extraction of solar photovoltaic models using an improved whale optimization algorithm, *Energy Convers. Manag.* 174 (2018) 388–405.
- [121] H.M. Hasanien, Performance improvement of photovoltaic power systems using an optimal control strategy based on whale optimization algorithm, *Electr. Power Syst. Res.* 157 (2018) 168–176.
- [122] B. Mahdad, Improvement optimal power flow solution under loading margin stability using new partitioning whale algorithm, *Int. J. Manag. Sci. Eng. Manag.* (2018) 1–15.
- [123] I.R. Kumawat, S.J. Nanda, R.K. Maddila, Positioning LED panel for uniform illuminance in indoor VLC system using whale optimization, *Optical and Wireless Technologies* 472 (2018) 131–139.
- [124] A. Kumar, V. Bhalla, P. Kumar, T. Bhardwaj, N. Jangir, Whale optimization algorithm for constrained economic load dispatch problems-A cost optimization, *Ambient Communications and Computer Systems* 696 (2018) 353–366.
- [125] S. Das, A. Bhattacharya, A.K. Chakraborty, V. Pandey, Fixed head short-term hydrothermal scheduling using whale optimization algorithm considering the uncertainty of solar power, in: *Ninth International Conference on Advanced Computing (ICoAC)*, IEEE, 2017, pp. 179–185.
- [126] O.S. Elazab, H.M. Hasanien, M.A. Elgendy, A.M. Abdeen, Whale optimisation algorithm for photovoltaic model identification, in: *The 6th International Conference on Renewable Power Generation (RPG)*, 2017, pp. 1906–1911. Issue 13.
- [127] S.H. Qazi, M.W.B. Mustafa, S. Soomro, R.M. Larik, An optimal current controller for photovoltaic system based three phase grid using whales optimization algorithm, in: *IEEE Conference on Energy Conversion (CENCON)*, 2017, pp. 15–20.
- [128] A. Rosyadi, O. Penangsang, A. Soeprijanto, Optimal filter placement and sizing in radial distribution system using whale optimization algorithm, in: *International Seminar on Intelligent Technology and its Applications (ISITIA)*, IEEE, 2017, pp. 87–92.
- [129] B.C. Neagu, O. Ivanov, M. Gavrila, Voltage profile improvement in distribution networks using the whale optimization algorithm, in: *9th International Conference on Electronics, Computers and Artificial Intelligence (ECAI)*, IEEE, 2017, pp. 1–6.
- [130] A. Saha, L.C. Saikia, Utilisation of Ultra-capacitor in Load Frequency Control under Restructured STPP-Thermal Power Systems Using WOA Optimised PIDNFPD Controller, *IET Generation, Transmission & Distribution*, 2017, pp. 3318–3331.
- [131] A. Marimuthu, K. Gnanambal, R. Priyanka, Optimal allocation and sizing of DG in a radial distribution system using whale optimization algorithm, in: *International Conference on Innovations in Green Energy and Healthcare Technologies (IGEHT)*, IEEE, 2017, pp. 1–5.
- [132] P.R. Sahu, P.K. Hota, S. Panda, Power system stability enhancement by fractional order multi input SSSC based controller employing whale optimization algorithm, *J. Elec. Syst. Inf. Technol.* 5 (3) (December 2018) 326–336.
- [133] S. Raj, B. Bhattacharyya, Optimal placement of TCSC and SVC for reactive power planning using Whale optimization algorithm, *Swarm and Evolutionary Computation* 40 (2017) 131–143.
- [134] K.B.O. Medani, S. Sayah, A. Bekrar, Whale optimization algorithm based optimal reactive power dispatch: a case study of the Algerian power system, *Electr. Power Syst. Res.* 163 (2017) 696–705. Part B.
- [135] P.D.P. Reddy, V.C.V. Reddy, T.G. Manohar, Optimal renewable resources placement in distribution networks by combined power loss index and whale optimization algorithms, *Journal of Electrical Systems and Information Technology* 5 (2) (2017) 175–191.
- [136] M.N. Heris, M. Mehdinejad, B. Mohammadi-Ivatloo, G. Babamalek-Gharehpetian, Combined heat and power economic dispatch problem solution by implementation of whale optimization method, *Neural Comput. Appl.* (2017) 1–16.
- [137] P.D.P. Reddy, V.C.V. Reddy, T.G. Manohar, Whale optimization algorithm for optimal sizing of renewable resources for loss reduction in distribution systems, *Renew. Wind Water Solar* 4 (3) (2017) 1–13.
- [138] I.N. Trivedi, M. Bhoye, R.H. Bhesdadiya, P. Jangir, N. Jangir, A. Kumar, An emission constraint environment dispatch problem solution with microgrid using Whale Optimization Algorithm, in: *National Power Systems Conference (NPSC)*, IEEE, 2016, pp. 1–6.
- [139] I.N. Trivedi, N. Jangir, P. Jangir, M.H. Pandya, R.H. Bhesdadiya, A. Kumar, Price penalty factors based approach for emission constrained economic dispatch problem solution using whale optimization algorithm, in: *2016 IEEE 1st International Conference on Power Electronics, Intelligent Control and Energy Systems (ICPEICES)*, 2016, pp. 1–5.
- [140] D.B. Prakash, C. Lakshminarayana, Optimal siting of capacitors in radial distribution network using Whale Optimization Algorithm, *Alexandria Engineering Journal* 56 (4) (2016) 499–509.
- [141] Y. Moodi, S.R. Mousavi, A. Ghavidel, M.R. Sohrabi, M. Rashki, Using Response Surface Methodology and providing a modified model using whale algorithm for estimating the compressive strength of columns confined with FRP sheets, *Constr. Build. Mater.* 183 (2018) 163–170.
- [142] A. Kaveh, Sizing Optimization of Skeletal Structures Using the Enhanced Whale Optimization Algorithm, *Applications of Metaheuristic Optimization Algorithms in Civil Engineering*, 2017, pp. 47–69.
- [143] A. Kaveh, M.I. Ghazaan, Enhanced Whale Optimization Algorithm for Sizing Optimization of Skeletal Structures, *Mechanics Based Design of Structures and Machines*, Taylor & Francis, 2016, pp. 1–46.
- [144] M. Rohani, G. Shafabakhsh, A. Haddad, E. Asnaashari, The Workflow Planning of Construction Sites Using Whale Optimization Algorithm (WOA), *The Turkish Online Journal of Design, Art and Communication-TOJDAC*, 2016, pp. 2938–2951.
- [145] G.M.A. Soliman, T.H.M. Abou-El-Enien, E. Emary, M.M.H. Khorshid, A hybrid whale optimization algorithm with adaptive spiral for terrorism prediction (the case of Egypt), *European Journal of Scientific Research*, *European Journal of Scientific Research* 149 (2) (2018) 165–184.
- [146] M. Mafarja, S. Mirjalili, Whale optimization approaches for wrapper feature selection, *Appl. Soft Comput.* 62 (2018) 441–453.
- [147] G.I. Sayed, A. Darwish, A.E. Hassanien, J.S. Pan, Breast cancer diagnosis approach based on meta-heuristic optimization algorithm inspired by the bubble-net hunting strategy of whales, in: *International Conference on Genetic and Evolutionary Computing, ICGEC 2016: Genetic and Evolutionary Computing*, vol. 536, 2017, pp. 306–313.
- [148] R.K. Saidala, N.R. Devarakonda, Bubble-net hunting strategy of whales based optimized feature selection for e-mail classification, in: *2nd International Conference for Convergence in Technology (I2CT)*, IEEE, 2017, pp. 626–631.
- [149] M. Sharawi, H.M. Zawbaa, E. Emary, Hossam M. Zawbaa, E. Emary, Feature selection approach based on whale optimization algorithm, in: *Ninth International Conference on Advanced Computational Intelligence (ICACI)*, IEEE, 2017, pp. 163–168.
- [150] H. Arslan, M. Toz, Data clustering based on FCM and WOA, in: *26th Signal Processing and Communications Applications Conference (SIU)*, IEEE, 2018, pp. 1–4.
- [151] M. Canayaz, R. Ozdag, Data clustering based on the whale optimization, *Middle East Journal of Technic* 2 (1) (2017) 178–187.
- [152] M.P.K. Reddy, M.R. Babu, Implementing self adaptiveness in whale optimization for cluster head selection in Internet of Things, *Clust. Comput.* (2017) 1–12.
- [153] J. Nasiri, F.M. Khiyabani, A whale optimization algorithm (WOA) approach for clustering, *Journal of Cogent Mathematics & Statistics* 5 (1) (2017) 1–13.
- [154] A. Mostafa, A.E. Hassanien, M. Houseni, H. Hefny, Liver segmentation in MRI images based on whale optimization algorithm, *Multimed. Tool. Appl.* 76 (23) (2017) 24931–24954.
- [155] G. Hassan, A.E. Hassanien, Retinal fundus vasculature multilevel segmentation using whale optimization algorithm, *Signal, Image and Video Processing* 12 (2) (2017) 263–270.
- [156] S.J. Mousavirad, H. Ebrahimpour-Komleh, Multilevel image thresholding using entropy of histogram and recently developed population-based metaheuristic algorithms, *Evolutionary Intelligence* 10 (1–2) (2017) 45–75.
- [157] M.A. Algabalawy, A.Y. Abdelaziz, S.F. Mekhamer, S.H.E.A. Aleem, Considerations on optimal design of hybrid power generation systems using whale and sine cosine optimization algorithms, *J. Elect. Syst. Inf. Technol.* 5 (3) (December 2018) 312–325.
- [158] A. Srivastava, D.K. Das, A. Rai, R. Raj, Parameter estimation of a permanent magnet synchronous motor using whale optimization algorithm, in: *Recent Advances on Engineering, Technology and Computational Sciences (RAETCS)*, IEEE, 2018, pp. 1–6.
- [159] H.M. Hasanien, Whale optimisation algorithm for automatic generation control of interconnected modern power systems including renewable energy sources, *IET Gener., Transm. Distrib.* 12 (3) (2017) 607–614.
- [160] A.S. Ahmed, M.A. Attia, N.M. Hamed, A.Y. Abdelaziz, Comparison between genetic algorithm and whale optimization algorithm in fault location estimation in power systems, in: *Nineteenth International Middle East Power Systems Conference (MEPCON)*, IEEE, 2017, pp. 631–637.
- [161] X. Zhang, Z. Liu, Q. Miao, L. Wang, Bearing fault diagnosis using a whale optimization algorithm-optimized orthogonal matching pursuit with a combined time-frequency atom dictionary, *Mech. Syst. Signal Process.* 107 (2018) 29–42.
- [162] H.H. Mehne, S. Mirjalili, A parallel numerical method for solving optimal control problems based on whale optimization algorithm, *Knowl. Base Syst.* 151 (2018) 114–123.
- [163] J. Ruiye, C. Tao, W. Songyan, Y. Ming, Order whale optimization algorithm in rendezvous orbit design, in: *Tenth International Conference on Advanced Computational Intelligence (ICACI)*, IEEE, 2018, pp. 97–102.
- [164] X. Huang, R. Wang, X. Zhao, K. Hu, Aero-engine performance optimization based on whale optimization algorithm, in: *2017 36th Chinese Control Conference (CCC)*, 2017, pp. 11437–11441.
- [165] T.K. Dao, T.S. Pan, J.S. Pan, A multi-objective optimal mobile robot path planning based on whale optimization algorithm, in: *IEEE 13th International Conference on Signal Processing (ICSP)*, 2016, pp. 337–342.
- [166] M.M. Ahmed, E.H. Houssein, A.E. Hassanien, A. Taha, E. Hassanien, in: *International Conference on Advanced Intelligent Systems and Informatics, AISI 2017: Proceedings of the International Conference on Advanced Intelligent Systems and Informatics*, 2018, pp. 724–733.
- [167] M.F. Horng, T.K. Dao, C.S. Shieh, T.T. Nguyen, A Multi-Objective Optimal Vehicle Fuel Consumption Based on Whale Optimization Algorithm, *Advances in Intelligent Information Hiding and Multimedia Signal Processing*, 2017, pp. 371–380.
- [168] A.R. Jadhav, T. Shankar, Whale optimization based energy-efficient cluster head selection algorithm for wireless sensor networks, *Neural and Evolutionary Computing* (2017) 1–22.
- [169] S. Khalilpourazari, S.H.R. Pasandideh, A. Ghodratnama, Robust possibilistic programming for multi-item EOQ model with defective supply batches: whale Optimization and Water Cycle Algorithms, *Neural Comput. Appl.* (2018) 1–28.
- [170] G.N. Reddy, S.P. Kumar, Multi objective task scheduling algorithm for cloud computing using whale optimization technique, in: *International Conference on Next Generation Computing Technologies, NGCT 2017: Smart and Innovative Trends in Next Generation Computing Technologies*, vol. 827, 2018, pp. 286–297.

- [171] K. Sreenu, M. Sreelatha, W-Scheduler: whale optimization for task scheduling in cloud computing, *Clust. Comput.* (2017) 1–12.
- [172] M. Sharma, R. Garg, Energy-aware whale-optimized task scheduler in cloud computing, in: *International Conference on Intelligent Sustainable Systems (ICISS)*, 2017, pp. 121–126.
- [173] M. Fu, H. Zhonghua, G. Zhijun, D. Xiaoting, T. Xutian, Whale optimization algorithm for flexible flow shop scheduling with setup times, in: *9th International Conference on Modelling, Identification and Control (ICMIC)*, 2017, pp. 157–162.
- [174] C. Zhang, X. Fu, S. Peng, Y. Wang, Linear unequally spaced array synthesis for sidelobe suppression with different aperture constraints using whale optimization algorithm, in: *13th IEEE Conference on Industrial Electronics and Applications (ICIEA)*, 2018, pp. 69–73.
- [175] M. Abdel-Basset, D. El-Shahat, I. El-henawy, A.K. Sangaiah, S.H. Ahmed, A novel whale optimization algorithm for cryptanalysis in merkle-hellman cryptosystem, *Mobile Network. Appl.* 23 (4) (2018) 723–733.
- [176] A. Alzaqebah, R. Masadeh, A. Hudaib, Whale optimization algorithm for requirements prioritization, in: *9th International Conference on Information and Communication Systems (ICICS)*, 2018, pp. 84–89.
- [177] R.K. Saidala, N. Devarakonda, Improved whale optimization algorithm case study: clinical data of anaemic pregnant woman, *Data Engineering and Intelligent Computing* (2018) 271–281.
- [178] P. Sarkar, N.M. Laskar, S. Nath, K.L. Baishnab, S. Chanda, Offset voltage minimization based circuit sizing of CMOS operational amplifier using whale optimization algorithm, *J. Inf. Optim. Sci.* (2017) 1–17.
- [179] B. Zeng, L. Gao, X. Li, Whale swarm algorithm for function optimization, in: *International Conference on Intelligent Computing, ICIC 2017: Intelligent Computing Theories and Application*, 2017, pp. 624–639.
- [180] A. Merzi, V. Gazi, Chemical concentration map building using whale optimization algorithm, in: *IEEE Symposium Series on Computational Intelligence (SSCI)*, 2017, pp. 1–8.
- [181] R. Barham, I. Aljarah, Link prediction based on whale optimization algorithm, in: *International Conference on New Trends in Computing Sciences (ICTCS)*, 2017, pp. 55–60.
- [182] A. Mukherjee, N. Chakraborty, B.K. Das, Whale Optimization Algorithm: an Implementation to Design Low-Pass FIR Filter, *Innovations in Power and Advanced Computing Technologies (I-PACT)*, 2017, pp. 1–5.
- [183] I.N. Trivedi, J. Pradeep, J. Narottam, K. Arvind, L. Dilip, A Novel Adaptive Whale Optimization Algorithm for Global Optimization, Vol 9, 2016, pp. 1–6 (38).
- [184] M. Wozniak, D. Polap, On manipulation of initial population search space in heuristic algorithm through the use of parallel processing approach, in: *IEEE Symposium Series on Computational Intelligence (SSCI)*, 2016, pp. 1–6.
- [185] H. Hu, Y. Bai, T. Xu, A whale optimization algorithm with inertia weight, *WSEAS Trans. Comput.* 15 (2016) 319–326.
- [186] K.R. Opara, J. Arabas, Differential Evolution: a survey of theoretical analyses, *Swarm and Evolutionary Computation* 44 (2019) 546–558.
- [187] S. Das, S.S. Mullick, P.N. Suganthan, Recent advances in differential evolution – an updated survey, *Swarm and Evolutionary Computation* 27 (2016) 1–30.
- [188] A.P. Piotrowski, Review of differential evolution population size, *Swarm and Evolutionary Computation* 33 (Feb 2017) 1–24.
- [189] Z.Z. Liu, Y. Wang, S. Yang, K. Tang, An adaptive framework to tune the coordinate systems in nature-inspired optimization algorithms, *IEEE Transactions on Cybernetics* 49 (4) (2019) 1403–1416.
- [190] M. Mavrovouniotis, C. Li, S. Yang, A survey of swarm intelligence for dynamic optimization: algorithms and applications, *Swarm and Evolutionary Computation* 33 (2017) 1–17.
- [191] N. Lynn, M.Z. Ali, P.N. Suganthan, Population topologies for particle swarm optimization and differential evolution, *Swarm and Evolutionary Computation* 39 (2018) 24–35.
- [192] Y. Jin, Surrogate-assisted evolutionary computation: recent advances and future challenges, *Swarm and Evolutionary Computation* 1 (2) (2011) 61–70.
- [193] E. Hancer, D. Karaboga, A comprehensive survey of traditional, merge-split and evolutionary approaches proposed for determination of cluster number, *Swarm and Evolutionary Computation* 32 (2017) 49–67.
- [194] R. Malhotra, M. Khanna, R.R. Raje, On the application of search-based techniques for software engineering predictive modeling: a systematic review and future directions, *Swarm and Evolutionary Computation* 32 (2017) 85–109.
- [195] B.Y. Qu, Y.S. Zhu, Y.C. Jiao, M.Y. Wu, J.J. Liang, A survey on multi-objective evolutionary algorithms for the solution of the environmental/economic dispatch problems, *Swarm and Evolutionary Computation* 38 (2018) 1–11.
- [196] A. Zhou, B.-Y. Qu, H. Li, S.-Z. Zhao, P.N. Suganthan, Q. Zhang, Multiobjective evolutionary algorithms: a survey of the state-of-the-art, *Swarm and Evolutionary Computation* 1 (1) (2011) 32–49.
- [197] S. Mahdavi, S. Rahnamayan, K. Deb, Opposition based learning: a literature review, *Swarm and Evolutionary Computation* 39 (2018) 1–23.
- [198] P. Rakshit, A. Konar, S. Das, Noisy evolutionary optimization algorithms-A comprehensive survey, *Swarm and Evolutionary Computation* 33 (2017) 18–45.
- [199] A. Gotmare, S.S. Bhattacharjee, R. Patidar, N.V. Georg, Swarm and evolutionary computing algorithms for system identification and filter design: a comprehensive review, *Swarm and Evolutionary Computation* 32 (2017) 68–84.
- [200] A.A. Mousa, M.A. El-Shorbagy, W.F. Abd-El-Wahed, Local search based hybrid particle swarm optimization algorithm for multiobjective optimization, *Swarm and Evolutionary Computation* 3 (2012) 1–14.